
EGOSTREAM: A Diagnostic Benchmark for Streaming Episodic Memory in Egocentric Vision

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Abstract

Continuous episodic memory is a core capability for autonomous agents operating in dynamic, real-world environments, yet current streaming video benchmarks provide limited tools for diagnosing what models remember and for how long. We introduce EGOSTREAM, a diagnostic benchmark for streaming episodic memory evaluation in egocentric vision. EGOSTREAM organizes 2,250 curated questions along seven cognitive dimensions: detail, spatial, temporal, event, social, causal, and prospective memory. We introduce the Answer Validity Window (AVW), which specifies the temporal span an answer remains valid as the observed scene evolves. This allows us to expand the questions into 8,528 recall-conditioned evaluations, enabling controlled testing from instant to ultra-long-term recall while separating genuine model forgetting from natural world-state changes. We rigorously establish baseline performance through a unified streaming MLLM framework that compares several state-of-the-art memory-management mechanisms, covering sliding windows, attention sinks, KV-cache pruning, merging, and offloading. Experiments within a unified Qwen3-VL backbone reveal that comparable aggregate accuracies mask starkly different memory profiles. For instance, token pruning preserves fine-grained details and temporal structure significantly better than token merging, while quantized offloading rescues ultra-long-term recall. Ultimately, all mechanisms operate well below real-time (>1 s per frame), and top performing methods ceil at about 45% accuracy, exposing critical gaps in current architectures. EGOSTREAM provides the diagnostic testbed needed to close these gaps. **Project website, news and updates at:** <https://saroo25.github.io/Egostream/>

1 Introduction

To operate successfully in the physical world, an autonomous agent must store and recall episodic events across both short and long temporal horizons [41, 40, 8]. For visual embodied AI systems, streaming egocentric video offers a natural testbed for studying continuous memory, as it captures first-person experience as it unfolds over time [17, 3], unlike detached exocentric media such as movies or surveillance footage. This makes the egocentric domain ideal for real-world memory tasks, including wearable assistants and embodied agents that must provide context-aware support [17, 33, 47, 24, 32]. In this setting, egocentric assistants must answer questions about past events (e.g., whether the user locked the door or where an object was placed), processing observations incrementally and maintaining a compact memory under bounded resources. This makes memory in egocentric vision particularly challenging: relevant evidence may appear long before it is queried, requiring the model to decide what to retain during streaming perception.

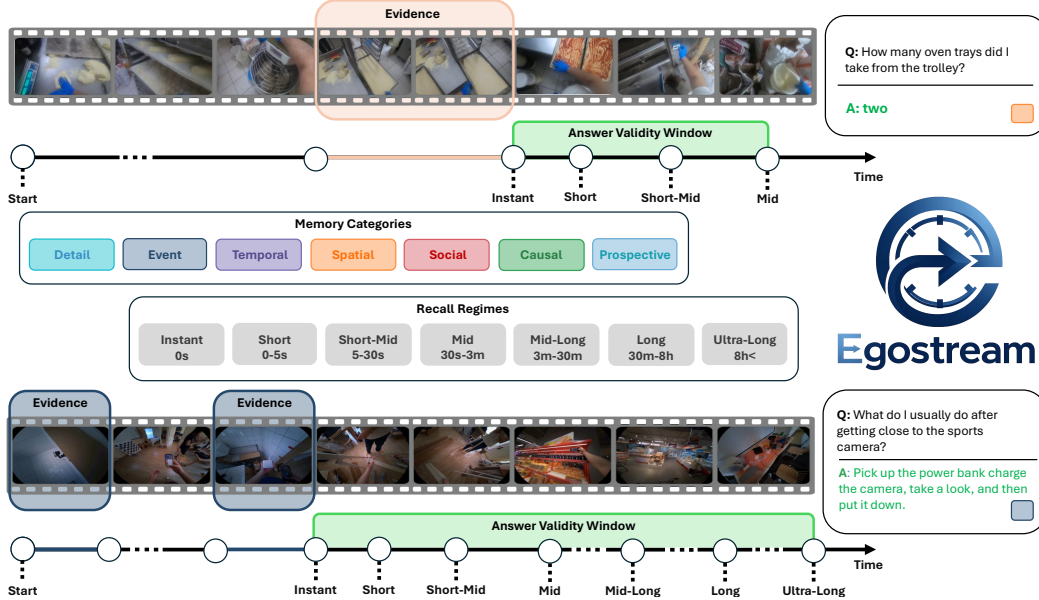


Figure 1: **Overview of the EGOSTREAM benchmark.** Each question is grounded in specific visual evidence, assigned a semantic category, and evaluated across multiple recall regimes. We introduce the Answer Validity Window as the exact temporal span during which an answer remains factually correct, constraining recall evaluations to this window to ensure consistency.

Despite recent progress in egocentric Video Question Answering (VideoQA), existing offline benchmarks still provide only a limited view of episodic memory [17, 24, 10, 6, 34, 32]. While they have substantially advanced long-form understanding and reasoning, they do not isolate memory as a measurable variable. The temporal distance between the supporting evidence and the query is rarely controlled, and evaluation does not distinguish *what* information must be remembered, *how long* it must be retained, or *when* a previously correct answer becomes invalid in a changing environment. Recent online and streaming benchmarks move closer to this setting [22, 31, 45, 42], but primarily measure online responsiveness or aggregate accuracy rather than providing a diagnostic account of memory quality. Consequently, models with very different memory profiles may obtain similar scores. One may preserve coarse event gist while forgetting object locations, whereas another may retain short-term spatial details but fail at long-horizon causal recall. This flattened assessment makes it difficult to determine which forms of episodic information are preserved by current streaming models and which degrade first. This limitation has led to a proliferation of memory-management methods that address streaming constraints [46, 53, 23, 21, 4, 48, 20]. Evaluating these methods solely through aggregate performance masks the qualitative trade-offs induced by their architectural choices. We argue that streaming memory evaluation should measure not only whether a model answers correctly, but also what kind of episodic information is retained and for how long.

To support progress in visual episodic memory systems, we introduce EGOSTREAM (Figure 1), a new benchmark curated from existing egocentric VideoQA datasets [10, 6, 34, 32, 47] and reorganized through a cognitive- and application-oriented framework. Drawing inspiration from cognitive accounts of episodic memory and related memory functions [41, 40, 5, 15, 35, 39, 26], we curate queries to test seven dimensions of memory: *detail*, *spatial*, *temporal*, *event*, *social*, *causal*, and *prospective* memory. To support a consistent evaluation of memory retention over time, we introduce the concept of *Answer Validity Window* (AVW)—a temporal span indicating the timeframe during which a specific past observation remains factually true before being overturned by subsequent events. Because continuous egocentric vision is dynamic, annotating AVWs ensures that forgetting is not conflated with scene evolution. Each question is then evaluated across *instant recall*, *short-term memory*, *long-term memory*, and *ultra-long-term memory*. This organization is also consistent with classic studies showing that memory should be evaluated across controlled retention intervals [13, 30, 12, 28].

Table 1: EGOSTREAM versus related egocentric and streaming VideoQA benchmarks. $\triangle$ denotes that a property is partially covered, but not in an explicit, controlled, and systematic form.

Benchmark	Egocentric	Online / streaming	Evidence grounding	Memory taxonomy	Answer Validity Window	Multiple recall regimes per QA
MovieChat-1K [36]	$\times$	$\times$	$\times$	$\triangle$	$\times$	$\times$
StreamingBench [22]	$\times$	$\checkmark$	$\times$	$\times$	$\times$	$\times$
EgoSchema [24]	$\checkmark$	$\times$	$\times$	$\times$	$\times$	$\times$
MM-Ego [49]	$\checkmark$	$\times$	$\times$	$\triangle$	$\times$	$\times$
EgoTextVQA [54]	$\checkmark$	$\checkmark$	$\times$	$\times$	$\times$	$\times$
EgoTempo [34]	$\checkmark$	$\times$	$\triangle$	$\triangle$	$\times$	$\times$
OVO-Bench [31]	$\triangle$	$\checkmark$	$\times$	$\triangle$	$\times$	$\times$
GroundVQA [10]	$\checkmark$	$\times$	$\checkmark$	$\times$	$\times$	$\times$
Multi-Hop EgoQA [6]	$\checkmark$	$\times$	$\checkmark$	$\times$	$\times$	$\times$
HD-EPIC [32]	$\checkmark$	$\times$	$\checkmark$	$\times$	$\times$	$\times$
EgoLifeQA [47]	$\checkmark$	$\times$	$\checkmark$	$\triangle$	$\times$	$\times$
EGOSTREAM (ours)	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$

To establish a picture of current progress we implement core memory-management strategies (e.g., sliding windows, attention sinks, token pruning, token merging) within a unified MLLM architecture. This controlled setup reveals that similar aggregate accuracy can hide distinct memory profiles, with policies preserving different semantic information and degrading differently across recall horizons. Our experiments reveal that established state-of-the-art memory management mechanisms exhibit only marginal performance differences when evaluated within our unified framework and that aggregate accuracies mask divergent memory profiles. For instance, token pruning preserves fine-grained details and temporal structure significantly better than token merging. This highlights the need to evaluate streaming memory by the specific episodic information preserved and its temporal accessibility, rather than just average accuracy.

Our contributions are threefold. 1) We introduce EGOSTREAM, a benchmark for streaming episodic memory in egocentric vision, built from five egocentric datasets and comprising 2,250 validated memory questions expanded into 8,528 recall-conditioned evaluations. 2) We organize episodic memory along seven semantic dimensions and introduce the Answer Validity Window (AVW) to measure temporal retention as the world evolves. 3) We provide a unified streaming MLLM framework for the controlled comparison of memory-management techniques. Annotations, code, and evaluation scripts are shared through the provided anonymous GitHub repository.

2 Related Work

Episodic Memory in Egocentric Vision Episodic memory in egocentric vision initially focused on offline temporal localization [17] and grounded QA [3, 10, 19]. Recognizing the continuous nature of first-person experience, recent efforts have shifted toward online streaming. While some approaches track memory through object-centric representations [25, 16, 14], VideoQA emerge as a suitable protocol for episodic retrieval, accelerated by MLLMs [10]. This shift has driven diverse continuous memory strategies, including dynamic KV-cache retrieval [11], cache compression [20, 7], and hierarchical memory reuse [51, 50, 46]. However, these works treat streaming memory as a modeling challenge, evaluated implicitly through end-task VideoQA. Repeated stream queries directly probe whether an observed fact is retained, forgotten, or superseded. EGOSTREAM builds on this by making episodic recall the primary evaluation target, not merely a means for downstream VideoQA.

VideoQA and Streaming Benchmarks. Existing benchmarks have significantly advanced long-form video comprehension [36], with recent egocentric datasets specializing this setting for first-person reasoning. These works cover a broad spectrum of abilities, including long-form QA [24, 49, 54], temporal grounding and multi-step reasoning [10, 6, 34], and fine-grained daily-life assistance [32, 47]. In parallel, streaming-oriented benchmarks like StreamingBench [22] and OVO-Bench [31] evaluate online responsiveness by posing queries during continuous video streams rather than assuming full offline access. Despite this progress, memory remains an implicit byproduct of broader video understanding or online interaction. As shown in Table 1, prior benchmarks address aspects like egocentric perception, streaming evaluation, or temporal reasoning. However, none jointly provide an explicit memory taxonomy, Answer Validity Windows, and controlled recall regimes. EGOSTREAM addresses this gap by turning streaming episodic memory into a specific, diagnostic evaluation target.

Table 2: Taxonomy of episodic memory dimensions, cognitive probes, and example queries.

Category	Cognitive Probe	Focus	Example Query
Detail	<i>What</i>	Object identity, attributes, counts	<i>“What did I put in the microwave?”</i>
Spatial	<i>Where</i>	Location, placement, spatial relations	<i>“Where did I put the pair of scissors?”</i>
Temporal	<i>When</i>	Order, sequence, time-indexed states	<i>“Where was the knife before I picked it?”</i>
Event	<i>How / Occurred</i>	Action gist, execution, state changes	<i>“Did I close the circuit breaker box?”</i>
Social	<i>Who</i>	People, interactions, co-presence	<i>“Who was with me when I ate?”</i>
Causal	<i>Why</i>	Reasons, explanations for past events	<i>“Why didn’t we buy the chestnut kernels?”</i>
Prospective	<i>What next</i>	Future plans and intended actions	<i>“What should I buy to decorate my room?”</i>

Streaming Memory Architectures in Video LLMs. To bound memory growth in continuous video streams, recent Video-LLMs employ mechanisms such as recency-based retention [44], KV-cache pruning [53, 21, 20, 7], token merging [4, 52, 43], and external retrieval [11]. These methods reduce spatial and temporal redundancy [20, 7, 48], often aided by quantization [7]. While they improve latency and efficiency, they are typically evaluated only via aggregate online accuracy, leaving unclear what each policy preserves and how abilities decay over time. EGOSTREAM addresses this by evaluating representative memory families under a unified backbone, enabling controlled analysis of what streaming models remember, for how long, and which dimensions fail first.

3 The EGOSTREAM Benchmark

3.1 Source Curation and Initial Set of Questions

We curate question-answer sets from five egocentric VideoQA datasets: **Ego4D Episodic Memory VQA** [3], **EgoLife** [47], **EgoTempo** [34], **Multi-Hop EgoQA** [6], and **HD-EPIC** [32]. These sources cover complementary aspects of egocentric perception, including episodic memory and temporal localization, long-context life assistance, fine-grained temporal reasoning, multi-hop grounded reasoning, and kitchen activity recognition. To unify these annotations under a streaming evaluation protocol, we apply a standardized curation pipeline. First, we ensure that all queries are explicit retrospective memory probes. Questions from Ego4D [3], EgoLife [47], and EgoTempo [47] are already suitable, while those from HD-EPIC [32] and Multi-Hop EgoQA [6] are rewritten using Gemini 3.1 [38], e.g., *“What is the best description for why the person performed the action <tip bowl> in video 1?”* is rewritten into *“What was the person trying to achieve by tipping the bowl?”*

To support a standardized 4-way multiple-choice evaluation, we use Gemini 3.1 [38] to generate three plausible distractors for open-ended datasets, namely EgoTempo [34] and Multi-Hop EgoQA [6]. We then apply an LLM-based quality-control step to remove ambiguous, trivial, or overly similar negatives, while preserving the original multiple-choice options for the remaining datasets (see Appendix A.1). Finally, each question is attached the temporal segment containing the visual evidence needed to infer the correct answer, which we refer to as the *evidence moment* (see Appendix A.2). These annotations are inherited from the source datasets. This yields 2,634 QA sets, that undergo a comprehensive human validation step. Annotators manually verify the accuracy of the question-answer pairs and the plausibility of the distractors (See Appendix A.3). This step filters out roughly 12% of the initial pool, yielding 2,335 human-verified, temporally grounded video QA instances.

3.2 The Cognitive Dimensions of Episodic Memory

EGOSTREAM isolates *what* a model remembers by partitioning queries into seven dimensions grounded in foundational cognitive theory [41, 40]. We structure this taxonomy around the natural “W-question” framework of human recall. Building upon the core episodic triad of *what*, *where*, and *when* [8, 5, 15], we incorporate expanded dimensions critical for embodied agents: procedural actions (*how*) [35], social context (*who*) [18, 37], causal reasoning (*why*) [39], and prospective intentions (*what next*) [26]. These categories are defined and summarized with examples in Table 2. Because episodic queries frequently involve overlapping cognitive demands (for instance, *“Who was with me when I operated the machine?”* contains both social and temporal cues) we assign each question one primary label and zero or more secondary tags, following the taxonomy in Table 2. To efficiently scale this process, we employed an iterative Human-in-the-Loop (HITL) annotation pipeline (see Appendix A.4). After this process 85 questions were discarded for not aligning to any of the category

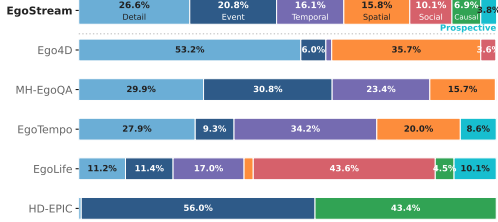


Figure 2: Distribution of question categories in EGOSTREAM and across subsets.

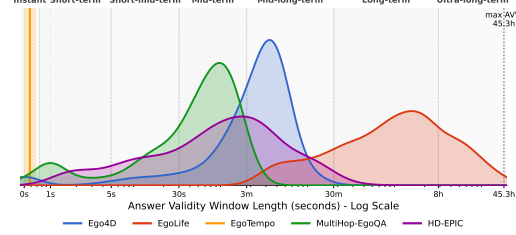


Figure 3: Recall regimes align with the empirical distribution of AVW lengths across datasets.

Table 3: Taxonomy of recall regimes aligned with memory consolidation timescales and distribution of memory categories that resolve within each temporal window.

Recall Regime	Window	Memory Mechanism	Psychological Basis	AVW Lengths Query Distribution
Instant	0s	Perceptual buffer	Encoding/immediate retrieval [9]	
Short-term	0s–5s	Working memory	Active maintenance [1]	
Short-mid-term	5s–30s	Working memory	Attention-dependent retention [1]	
Mid-term	30s–3m	Transitional	Early Synaptic Consolidation [12]	
Mid-long-term	3m–30m	Intermediate	Synaptic consolidation [27]	
Long-term	30m–8h	Long-term	Late Synaptic Consolidation [12]	
Ultra-long-term	> 8h	Long-term	Hippocampal-cortical transfer [28]	

■ Causal
 ■ Detail
 ■ Event
 ■ Prospective
 ■ Social
 ■ Spatial
 ■ Temporal

or being unclear, leaving 2, 250 valid questions (comprising 578 from Multi-Hop EgoQA [6], 498 from Ego4D [3], 433 from EgoLife [47], 425 from EgoTempo [34], and 316 from HD-EPIC [32]). Figure 2 reports the distribution of primary categories and across data subsets. For reference, the text-only baseline obtains an average accuracy of 25.42% demonstrating no stronger text bias See Appendix A.8 for a breakdown by semantic categories.

3.3 The Answer Validity Window and Recall Regimes

Time is the second fundamental dimension of episodic memory recall. To systematically evaluate this axis, we introduce the concept of *Answer Validity Window* (AVW), i.e., the precise temporal interval during which a question’s original answer remains semantically valid after all its supporting visually relevant evidence has been observed. Identifying this validity window is critical in continuous egocentric streams, where facts are inherently transient. An initially correct answer to a query such as “Is the glass full?” can rapidly become obsolete due to subsequent actions (e.g., drinking), object displacements (e.g., moving the glass), or broader scene evolutions. We annotate the AVW using the temporal structure of each source dataset. For EgoTempo, the AVW is 0, as questions are asked at the end of the trimmed input video. For Ego4D-VQA and Multi-Hop EgoQA, it spans from the end of the evidence moment to the end of the provided clip, while for EgoLife it extends to the long-term query time defined by the original dataset. For HD-EPIC, where validity intervals are not directly provided, we use Gemini 3.1 [38] to identify narration chunks in which the answer is observable and derive the AVW by trimming duplicate evidence around the ground-truth moment. All automatically derived annotations were manually reviewed for consistency. See Appendix A.5 for more details.

Associating an AVW to each question enables us to evaluate episodic memory at multiple recall regimes, reminiscent of classic memory testing paradigms [13, 30]. In practice, we sample multiple query times between the end of the relevant moment and the end of the validity interval, ensuring that forgetting is not conflated with answer invalidation caused by scene evolution. Drawing on established memory consolidation timescales [12, 28], application-oriented considerations [33], and empirical support in our data (see Figure 3), we define seven recall regimes spanning a log-scaled progression. Table 3 reports the proposed recall regimes and terminology, with supporting references from psychology and the empirical distribution of memory categories within each temporal window. Given the AVW annotations, we expand each curated question into one or more recall evaluations. Each question is first evaluated in the *instant* regime, immediately after the evidence moment, to measure whether the model can answer before memory decay becomes a factor. We then sample additional query times within the AVW according to our recall-regime taxonomy, with distribution parameters adjusted to match the empirical temporal structure of the source data. This produces a

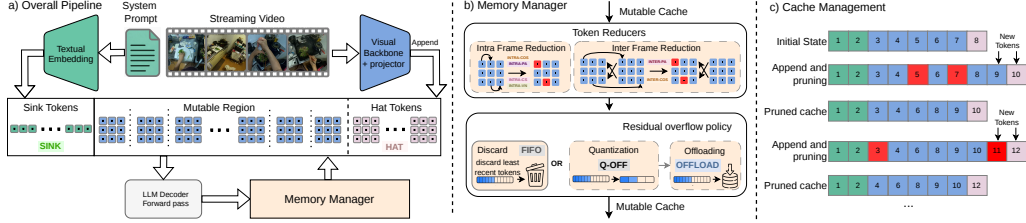


Figure 4: **Unified streaming framework.** The proposed framework processes frames sequentially while maintaining an incrementally updated KV cache. (a) The cache is partitioned into three logical regions: protected *sink tokens* (**SINK**), corresponding to the system prompt; a *mutable region*, containing historical visual tokens subject to memory management; and protected *hat tokens* (**HAT**), corresponding to the most recent visual observations. The pipeline incrementally appends new visual tokens to the cache, passes the updated cache to the LLM decoder, and calls the memory manager whenever the mutable cache exceeds its budget. (b) The memory manager operates only on the mutable cache. Optional reducer modules apply pruning or merging according to intra-frame, inter-frame, or global token-selection criteria. If the mutable cache still exceeds the live budget, an overflow policy either discards excess tokens or externalizes them through offloading or quantized offloading. (c) During the streaming inference, each new frame is encoded and appended to the cache. A fixed number of recent visual tokens enter the protected region, older tokens move into the mutable region, and the selected memory policy updates the mutable cache to satisfy the memory budget.

realistic set of recall delays across short- and long-horizon settings, yielding 8,528 question-recall evaluations. Further details are provided in Appendix A.6.

3.4 Evaluation Protocol and Metrics

To establish EGOSTREAM as a standardized reference for future researchers, we explicitly define inference rules and scoring metrics, as well as provide scripts to standardize evaluation.

Streaming Constraints. The benchmark strictly enforces a streaming inference protocol. Models must process the video sequentially: a query issued at timestamp t must be answered utilizing only the visual content observed up to t . Models are expected to maintain a compact internal representation of the past, avoiding to store raw video frames in an unbounded buffer to be reprocessed at query time. The visual processing required to form this memory must be purely query-agnostic, as the evaluation query is only revealed to the model at the exact moment of recall.

Evaluation. EGOSTREAM evaluates models by submitting the 2,250 valid questions across their applicable recall regimes, resulting in 8,528 individual inferences. Performance is assessed via standard accuracy (broken down across semantic categories and recall regimes) on the 4-way multiple-choice setup, with a chance-level baseline corresponding to 0.25. While individual methods are free to choose their specific system prompts, we utilize a unified prompt and answer-selection scheme for fair comparisons (see Appendix A.7). Methods are evaluated under limited memory budgets, and reporting the average processing time per frame.

4 A Unified Framework for Streaming Episodic Memory

Current evaluations of state-of-the-art models often confound memory mechanisms with orthogonal design choices, such as varying vision encoders or pre-training mixtures. To enable controlled diagnostic evaluations, we introduce a unified streaming Multimodal Large Language Model (MLLM) framework that isolates memory management as an independent variable. Instantiated on a frozen Qwen3-VL [2] backbone, our framework adapts to streaming VideoQA by continuously appending newly encoded visual tokens to the KV cache. This enables incremental video processing without recomputing past contexts at every query. By keeping the encoder, projector, LLM decoder, and prompting scheme strictly fixed, we provide a testbed for benchmarking distinct memory techniques. Our unified pipeline allows us to replicate popular state-of-the-art streaming video understanding approaches [7, 20, 48, 46, 11] on a plug-and-play memory manager that operates directly in KV-cache.

Figure 4 illustrates our unified streaming memory framework, detailing the cache organization, memory management, and update dynamics. The core principle of our approach is to restrict memory operations—such as merging, pruning, and offloading—strictly to historical visual evidence. This ensures that both the system prompt and the most recent observations are always preserved. As the video stream progresses, the memory manager continuously enforces a strict budget on the active cache. At question time, the model answers using this curated cache, drawing upon the most relevant available evidence selected from live blocks or, when retrieval is enabled, offloaded memory.

4.1 Memory Management Strategies

In the following, we detail the main Memory Management Strategies evaluated within our framework, which draw upon established techniques from prior KV-cache and streaming-video memory research. See Appendix B for more details.

Protected cache regions. Following attention-sink and streaming-inference methods, which show that structural tokens are critical for stable long-context decoding [46, 20, 53], we preserve the system prompt as fixed *sink tokens* (**SINK**). We also reserve a small protected region for the most recent visual tokens, denoted *hat tokens* (**HAT**), analogous to recent-token budgets in KV-cache reduction and streaming methods [53, 44, 20]. This organization (Figure 4a) preserves instruction-following and grounding for recent content, restricting memory management to the mutable historical cache.

Intra-frame reduction. Treating the mutable cache as an undifferentiated pool can discard temporal coverage. To prevent this, we evaluate frame-local reducers that compress redundant evidence within individual frames (Figure 4b). We apply pruning or merging based on four criteria: cosine redundancy (**INTRA-COS**) between adjacent key states; pseudo-attention salience (**INTRA-PA**) to discard low-attended tokens [7]; value-norm ranking (**INTRA-VN**) to prune low-norm states [20]; and sink-guided cross-attention (**INTRA-CS**) to remove tokens least attended by the system prompt [48].

Inter-frame reduction. Adjacent egocentric frames often share highly redundant content, enabling temporal compression. Following prior streaming-video methods, our memory manager compares neighboring frames in the mutable cache to prune redundant tokens when the budget is exceeded. We evaluate two cross-frame reducers: cosine redundancy (**INTER-COS**), which matches tokens using cosine similarity [7], and pseudo-attention salience (**INTER-PA**), which uses cross-frame pseudo-attention to estimate token importance [48] (Figure 4b).

Residual overflow policy. When the reduced cache still exceeds its budget, we apply an overflow policy (Figure 4b): either **FIFO**, which discards the oldest tokens [44], or *offloading* (**OFFLOAD**), which preserves excess chunks on disk for retrieval at question time via cosine similarity [11]. To minimize storage costs, we also evaluate an *int8 quantized offloading* variant (**Q-OFF**) [7].

5 Experiments and Results

We evaluate the capabilities of current streaming architectures on EGOSTREAM via a controlled diagnostic evaluation using our unified framework, an analysis of semantic profiles and temporal decay, and a comparison of state-of-the-art models. Configuration and hardware detailed in Appendix C.1.

5.1 Controlled Diagnostic Evaluation

Table 4 shows that while aggregate performance gaps among memory strategies are narrow, critical variations exist in *what* is retained and for *how long*. Established methods like ReKV [11] and StreamVLM [46] remain highly competitive, performing on par with recent approaches like InfiniPot-V [20]. No single method strictly dominates, showing that some are locally optimal for spatial or temporal retention (Rows 11, 18), while others (Rows 1, 11) balance accuracy and processing time. These trade-offs reveal key trends that can guide the design of future streaming architectures.

The role of **SINK and **HAT** tokens.** The naive Sliding Window (**FIFO**, Row 1) yields the lowest accuracy. Adding **SINK** tokens (StreamVLM [46], Row 2) boosts semantic and temporal averages from 29.53%/28.33% to 44.74%/39.05%, with negligible latency change (1.05s to 1.07s). This confirms that preserving system prompts is essential for stable inference [44]. Including **HAT** tokens adds a small benefit under the same policy, increasing semantic/temporal accuracy from 44.38%/40.23% to 45.19%/40.43% (Rows 10 and 17). These results show the value of protecting structural context.

Table 4: **Evaluation of memory-management strategies.** Accuracy is reported across semantic categories and temporal recall regimes, alongside macro-averages and per-frame processing time. Configurations are mapped to their closest equivalent in literature wrt isolated memory strategies.

Strategy	Configuration	Semantic Categories								Temporal Recall Regimes								Proc.
		Avg	SPAT	TEMP	DETL	EVNT	SOC	CAUS	PROS	Avg	Inst	Short	S-Mid	Mid	ML	Long	Ultra	(s)
Sliding-window baselines																		
1	FIFO	29.53	30.50	33.53	30.62	30.02	26.85	25.94	29.28	28.33	31.83	31.09	28.71	27.80	28.42	28.12	22.37	1.05
2	FIFO SINK	44.74	57.95	51.22	54.90	46.36	28.87	38.68	35.20	39.05	55.06	55.10	49.18	40.00	29.34	24.93	19.74	1.07
Similarity-based merge FIFO SINK HAT +																		
3	MERGE INTRA-PA	41.87	54.25	48.20	51.49	42.48	28.27	37.48	30.92	37.73	52.47	51.98	44.00	37.20	28.65	23.48	26.32	2.87
4	MERGE INTRA-COS	43.42	57.30	49.83	53.28	45.87	27.52	38.53	31.58	38.10	54.44	53.20	48.18	39.40	27.50	22.90	21.05	2.33
Similarity-based hybrid merge + prune FIFO SINK HAT +																		
5	MERGE INTRA-PA PRUNE INTER-PA	43.44	56.66	52.62	53.89	45.33	26.10	38.23	31.25	38.01	54.61	52.87	47.35	39.87	28.88	21.45	21.05	5.32
Similarity-based prune FIFO SINK HAT +																		
6	PRUNE INTRA-CS	42.39	52.25	51.80	52.10	43.85	27.23	37.93	31.58	38.88	52.12	50.64	45.82	39.07	28.42	25.80	30.26	4.91
7	PRUNE INTRA-VN	43.79	55.54	53.67	52.49	45.98	29.02	38.23	31.58	39.17	54.30	51.87	46.88	41.93	29.00	25.22	25.00	4.23
8	PRUNE INTER-COS	45.23	57.87	53.78	55.60	45.76	28.05	39.73	35.86	40.16	54.75	52.76	48.82	43.87	30.84	26.38	23.68	4.99
9	PRUNE INTRA-VN INTER-COS	44.52	58.91	56.23	55.51	46.36	26.10	39.58	28.95	40.22	54.44	53.93	50.12	42.27	30.38	21.45	28.95	4.42
10	PRUNE INTRA-PA INTER-COS	45.19	57.87	57.28	56.68	46.58	28.35	37.78	31.79	40.43	55.42	54.37	50.88	41.73	31.07	25.88	23.68	5.51
11	PRUNE INTRA-COS	45.63	60.51	52.39	56.61	46.64	28.50	39.88	34.87	39.34	55.60	54.99	52.18	41.27	30.61	24.93	15.79	2.28
12	PRUNE INTRA-PA	45.12	59.47	53.08	56.52	45.87	28.72	39.28	32.89	41.50	56.04	54.71	50.18	40.60	30.84	22.61	35.53	2.82
13	PRUNE INTRA-PA	45.42	59.87	55.53	56.17	45.65	27.30	38.53	34.87	40.30	55.42	54.32	49.35	42.80	30.84	24.35	25.00	4.82
14	PRUNE INTRA-PA INTER-PA	44.71	58.51	55.30	56.52	45.54	28.37	39.58	28.95	40.03	55.55	53.87	50.29	42.00	29.46	26.67	22.37	5.66
15	PRUNE INTRA-PA INTER-PA	44.42	57.62	56.46	56.21	46.75	26.85	39.43	27.63	39.91	55.10	54.04	50.06	42.13	30.15	22.90	25.00	4.67
16	PRUNE INTRA-CS INTER-PA	44.39	56.74	52.50	55.56	45.22	28.95	39.88	31.91	39.60	55.24	54.43	48.12	40.60	30.03	23.77	25.00	4.87
Similarity-based prune FIFO SINK +																		
17	PRUNE INTRA-PA INTER-COS	44.38	58.75	54.25	56.39	46.04	24.98	38.98	31.25	40.23	55.28	53.09	48.76	42.27	31.19	19.42	31.58	4.76
Offloading-based memory management FIFO SINK HAT																		
18	OFFLOAD	40.99	51.69	49.48	48.99	42.86	29.99	35.98	27.96	36.83	51.36	47.97	44.00	38.67	29.23	25.51	21.05	2.03
19	Q-OFF	41.63	51.85	50.64	50.83	43.63	27.52	35.38	31.58	38.77	52.61	47.74	43.12	39.27	30.72	26.38	31.58	1.59
20	PRUNE INTRA-PA INTER-COS Q-OFF	42.86	53.61	52.39	51.88	44.51	27.97	37.78	31.91	39.93	54.03	48.69	45.06	40.00	31.42	26.09	34.21	1.43

Memory budget: 9.804 tokens ($\approx 1.35GB$). *: variant without **Q-OFF**.

Merge vs. Prune. **PRUNE** mechanisms consistently outperform token **MERGE**. For **INTRA-PA**, pruning (Row 12) beats merging (Row 3) 45.12% to 41.87%. Discarding redundant tokens preserves episodic details better than averaging, which blurs fine-grained evidence (see DETL). While using a hybrid **PRUNE**/**MERGE** approach (Row 5) closes this gap (up to 45.97%), it drastically increases processing time (5.56s vs. ~ 2.5 s for pure intra-merging).

Optimal in-memory strategies. Combining intra- and inter-frame reduction yields the best streaming policies. The hybrid **PRUNE** **INTRA-PA** **INTER-COS** (Row 10) provides the strongest overall trade-off between semantic and temporal recall (45.19%/40.43%), excelling in Spatial and Detail memory. Blending local salience filtering with cross-frame redundancy checks optimally preserves visual anchors. However, this peak accuracy incurs the highest processing time (5.51s), more expensive than variants like **PRUNE** **INTRA-COS** (Row 11, 2.28s).

Offloading. **OFFLOAD** keeps evicted memory on disk, but alone is weaker than in-memory pruning. Row 18 reaches 40.99% / 36.83% with 2.0966 GB $\times$ min. **Q-OFF** halves storage to 1.0515 GB $\times$ min and improves accuracy (Row 19: 41.63% / 38.77%, 1.59s), while StreamingTOM (Row 20) obtains better results combining low-salience tokens **PRUNE** + **Q-OFF** for 42.86% / 39.93%, gaining 34.21% for Ultra-Long recall and only 0.4847 GB $\times$ min.

5.2 State-of-the-Art Streaming Models

When comparing state-of-the-art streaming architectures under our unified KV-cache framework, distinct accuracy–latency trade-offs emerge rather than a strict ranking (Table 4). **StreamVLM** [46] serves as the strongest low-latency baseline, proving that protecting **SINK** tokens yields high semantic (44.74%) and temporal (39.05%) accuracy with minimal overhead (1.07s). For tasks requiring higher fidelity, token pruning models like **StreamMem** [48], **InfiniPot-V** [20], and **StreamTOM** [7] (without offloading) set the benchmark’s upper limits—peaking at 45.23% semantic and 40.43% temporal accuracy—but suffer from severe processing delays (4.23s–5.51s). Token merging (**CaM-like** [52]) offers a middle ground, cutting pruning latency in half (2.33s–2.87s) while remaining competitive. Finally, external retrieval strategies like **ReKV** [11] and full **StreamingTOM** [7] reveal that quantized offloading is beneficial for long-horizon memory; it rescues ultra-long recall (reaching 34.21% for StreamingTOM) and slashes processing time (down to 1.43s), albeit with a slight dip in average accuracy. Ultimately, while current SOTA methods successfully specialize in different memory profiles, all still operate well below 1 FPS, highlighting that achieving true real-time streaming ingestion remains an open challenge.

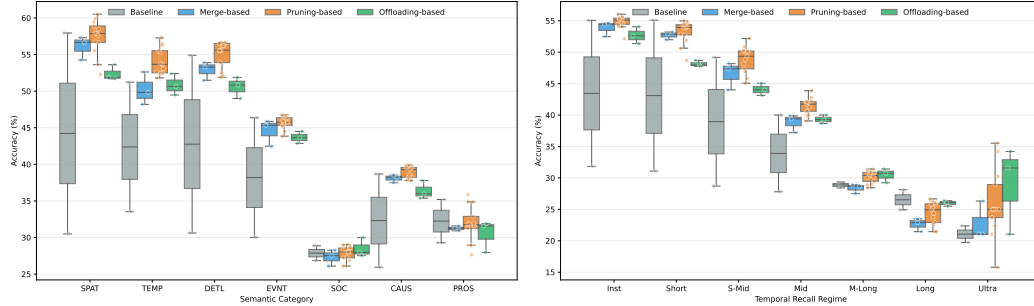


Figure 5: **Semantic and Temporal profiles.** Accuracy distributions of baselines (no **PRUNE**/**MERGE**), merge-, prune-, and offloading-based methods wrt semantic categories (left) and recall times (right).

Table 5: Policy **FIFO** **SINK** **HAT** **PRUNE** **INTRA-PA** **INTER-COS** under different memory budgets (tokens).

Tokens	Semantic Categories								Temporal Recall Regimes								Time (s)
	Avg	SPAT	TEMP	DETL	EVNT	SOC	CAUS	PROS	Avg	Inst	Short	S-Mid	Mid	ML	Long	Ultra	Proc.
1 2636	40.88	51.52	49.83	51.01	42.37	27.75	35.08	28.62	35.59	51.76	49.36	43.35	37.53	31.19	21.45	14.47	2.21
2 3660	42.43	54.65	50.99	52.84	44.07	26.93	34.63	32.89	38.80	52.25	50.97	45.06	39.47	31.30	24.93	27.63	2.78
3 5708	43.98	57.22	52.85	55.42	45.49	27.30	39.28	30.26	40.61	54.79	52.70	48.24	40.73	30.72	25.51	31.58	3.40
4 9804	45.19	57.87	57.28	56.68	46.58	28.35	37.78	31.79	40.43	55.42	54.37	50.88	41.73	31.07	25.88	23.68	5.51
5 17996	45.71	61.16	56.46	58.66	46.97	28.05	41.68	26.97	41.18	57.29	54.93	50.88	44.33	32.34	23.48	25.00	4.35
6 34380	46.25	62.44	57.04	59.14	45.95	26.78	40.39	32.00	41.11	56.62	54.48	52.24	44.43	32.14	23.88	23.94	5.89
7 50764	47.08	61.85	57.97	58.91	47.59	25.46	43.77	34.00	41.94	57.10	55.58	51.78	43.58	30.71	25.67	29.17	5.90

5.3 Semantic Profiles and Temporal Decay

Figure 5 shows that merging, pruning and offloading techniques have different effects across semantic categories and temporal recall regimes. While all improve over baselines, pruning-based methods often have advantages, especially in the temporal and detail categories, and offloading-based approaches achieve better results on social and in medium-long to ultra-long recall. Overall, these results highlight the importance of evaluating across semantic categories and recall times, rather than just by aggregate accuracies.

5.4 Memory Budget

Table 5 shows that memory budget is a driver of accuracy, but with diminishing returns. Increasing the budget from 2.6K to 9.8K tokens, from Row 1 to Row 2, raises semantic accuracy from 40.88% to 45.19% and temporal accuracy from 35.59% to 40.43%. Larger budgets further improve the averages, peaking at 47.08% semantic and 41.94% temporal with 50.8K tokens in Row 7, but nearly triple processing time relative to the smallest budget in Row 1.

6 Conclusion and Limitations

EGOSTREAM establishes a diagnostic benchmark and unified framework for evaluating streaming episodic memory in egocentric vision. By introducing the Answer Validity Window (AVW) to decouple genuine forgetting from natural scene changes, our evaluations reveal that comparable aggregate accuracies often mask distinct memory vulnerabilities. Moving beyond simple performance metrics, EGOSTREAM provides a precise tool to diagnose exactly what, and for how long, embodied systems remember—laying the groundwork for reliable, context-aware assistive agents.

Limitations and Societal Impact EGOSTREAM establishes a foundation benchmark for streaming episodic memory, but several limitations remain. Although we provide robust baselines, memory decay may vary across backbones and implementations. While EGOSTREAM can support open-ended QA, we focus on multiple choice, leaving free-form generation unexplored. Finally, EGOSTREAM inherits environmental and geographic biases from its source datasets and narrations. Streaming episodic memory may benefit assistive technologies, but also raises privacy and surveillance risks by capturing bystanders or sensitive interactions; these risks require mitigation before deployment.

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A The EGOSTREAM Benchmark

This appendix reports additional details for the EGOSTREAM Benchmark construction process.

A.1 Hard Negative Generation Protocol

To support a unified multiple-choice evaluation protocol, we generated three hard negative answers for each valid open-ended question-answer pair. The protocol is data-agnostic: given a question q and a ground-truth answer a , it produces a set of plausible but incorrect distractors that preserve the linguistic form, level of specificity, and granularity of the correct answer while changing its factual content. This design prevents multiple-choice evaluation from becoming artificially easy due to generic, implausible, or structurally mismatched alternatives, and encourages models to rely on episodic visual evidence rather than superficial answer-format cues.

We instantiated the generation and validation pipeline using Gemini 3.1 [38] as the LLM backend. For each valid question-answer pair, the model was queried in JSON-only mode. Before generation, we discarded items whose answer was empty or malformed. For instance, questions with placeholder answers such as “.” were removed from the generation pool, since they do not provide a reliable positive answer from which to construct structured distractors.

We then generated one initial negative answer conditioned on both the question and the ground-truth answer, using the Prompt 1. This first step allows the model to explicitly match the structure of the correct answer: if the ground truth is a short noun phrase, the negative is also required to be a short noun phrase; if the ground truth is a full sentence, the negative must preserve a comparable sentence structure. The prompt further instructs the model to keep similar length, tense, phrasing style, and level of detail, while changing the core content.

After this first negative was obtained, we generated the remaining two negatives without exposing the ground-truth answer to the model, using the Prompt 2. Instead, the model was given only the question and the first accepted negative answer. This second stage reduces the risk that all distractors are produced by directly perturbing the same ground-truth phrase, while still enforcing consistency in answer format. In other words, the first negative acts as a style and granularity anchor for the remaining distractors.

As an illustrative example, consider the following question:

What objects did I connect the white and black cable to?

with ground-truth answer:

You connected the cable to the charger and the laptop.

The generated negatives were:

You connected the cable to the external monitor and the docking station.

You connected the cable to the wireless router and the network switch.

You connected the cable to the external hard drive and the USB hub.

All three distractors preserve the same answer structure as the ground-truth answer, namely “You connected the cable to [object] and [object].” They also maintain comparable granularity by referring to plausible electronic devices and connection targets. However, the content is changed, since none of the distractors corresponds to the correct pair of objects connected in the video. This example illustrates the intended behavior of the pipeline: negatives should be difficult because they are stylistically and semantically plausible, not because they are vague or unrelated.

All candidate negatives were then passed through two levels of filtering. First, we applied lexical and string-similarity checks to reject candidates that were too close to the ground-truth answer or duplicated another already accepted negative. Answers were normalized by lowercasing, stripping punctuation and articles, and collapsing whitespace. We then computed lexical similarity using Python’s `difflib.SequenceMatcher` over normalized answer strings. The resulting score lies in $[0, 1]$ and is computed as $2M/T$, where M is the number of matching characters in the aligned matching blocks and T is the total number of characters in the two normalized strings. A score of 1

Algorithm 1: Hard negative generation and validation

Input: Question q , ground-truth answer a , target number of negatives $K = 3$

Output: Set of validated negatives $\mathcal{N}$

if a is empty or malformed **then**

 discard item

$\mathcal{N} \leftarrow \emptyset$

Generate first candidate $\tilde{a}_1$ from (q, a)

if ISVALIDNEGATIVE($q, a, \tilde{a}_1, \mathcal{N}$) **then**

$\mathcal{N} \leftarrow \mathcal{N} \cup \{\tilde{a}_1\}$

else

 retry first-negative generation up to a fixed limit

Generate additional candidates $\tilde{a}_2, \tilde{a}_3$ from $(q, \tilde{a}_1)$

foreach candidate $\tilde{a}$ in $\{\tilde{a}_2, \tilde{a}_3\}$ **do**

if ISVALIDNEGATIVE($q, a, \tilde{a}, \mathcal{N}$) **then**

$\mathcal{N} \leftarrow \mathcal{N} \cup \{\tilde{a}\}$

while $|\mathcal{N}| < K$ and regeneration budget not exhausted **do**

 Generate a new candidate $\tilde{a}$ from $(q, \tilde{a}_1, \mathcal{N} \cup \{a\})$

if ISVALIDNEGATIVE($q, a, \tilde{a}, \mathcal{N}$) **then**

$\mathcal{N} \leftarrow \mathcal{N} \cup \{\tilde{a}\}$

if $|\mathcal{N}| < K$ **then**

 discard item

return first K elements of $\mathcal{N}$

indicates identical normalized strings. A candidate was rejected if it was identical or near-identical to the ground-truth answer, using a threshold of 0.92, or if it duplicated an already accepted negative, using a threshold of 0.90.

Second, every remaining candidate was evaluated by an LLM validator (Gemini 3.1) using the Prompt 4. The validator was shown the question, the ground-truth answer, and the candidate negative, and was asked to verify that the candidate satisfies all required properties: factual content must differ from the ground truth, answer structure must be preserved, granularity must remain comparable, the answer must be plausible for the visual-question context, and the distractor must not be obviously easy. A candidate was accepted only if all validation flags were satisfied. If fewer than three candidates passed validation, we entered a regeneration loop in which additional candidates were generated one at a time using the question, the first accepted negative, and the list of banned answers, as specified in Prompt 3. This loop continued until three valid negatives were obtained or a maximum number of regeneration rounds was reached. Items for which three valid negatives could not be obtained were removed from the final multiple-choice subset.

Algorithm 1 summarizes the complete procedure.

The validation function ISVALIDNEGATIVE combines deterministic similarity filtering and model-based verification. Let $\text{norm}(\cdot)$ denote the normalized answer string and let $\text{sim}(\cdot, \cdot)$ denote sequence similarity. A candidate $\tilde{a}$ is rejected if $\text{norm}(\tilde{a}) = \text{norm}(a)$, if $\text{sim}(\tilde{a}, a) \geq 0.92$, or if it is identical or near-identical to any previously accepted negative, using a threshold of 0.90. Candidates passing these checks are then accepted only if the LLM validator confirms that the factual content changed, while structure, granularity, plausibility, and difficulty are preserved.

We applied this hard-negative generation protocol to the datasets in our benchmark pipeline that originally required multiple-choice distractor construction, namely MultiHop-EgoQA [6] and EgoTempo [34], while keeping all other data sources intact (Ego4D GroundVQA [10], EgoLife [47], and HD-EPIC [32]). After this first generation step, our dataset consists of 2,634 question-answer pairs with negatives.

Prompt 1: First negative generation

Generate exactly one incorrect answer for a visual QA example.

Goal:

The incorrect answer must be HARD:

- it must preserve the same answer structure as the ground-truth
↪ answer
- it must preserve the same level of detail/granularity
- it must preserve the same style and syntactic form
- but it must change the core content so that it is factually
↪ incorrect

Rules:

- If the ground-truth is a full sentence, output a full sentence with
↪ the same framing.
- If the ground-truth is a short noun phrase, output a short noun
↪ phrase.
- Keep similar length.
- Keep articles, tense, and phrasing style aligned with the
↪ ground-truth when possible.
- Change the object/entity/action detail so the answer is wrong.
- Do NOT make it obviously easy or generic.
- Do NOT explain.

Question: {question}

Ground truth answer: {gt_answer}

Return format:

```
{  
  "negative": "..."  
}
```

Prompt 2: Additional negative generation

Generate {n} additional incorrect answers for this question.

Important constraints:

- You are NOT given the ground-truth answer.
- Use only the question and the existing incorrect answer.
- The new answers must preserve the SAME structural style,
↪ granularity, and answer format
as the existing incorrect answer.
- They must be different in content from the existing incorrect
↪ answer and from each other.
- They must remain plausible and hard.
- Do NOT make them generic or obviously easy.
- Do not explain.

Question: {question}

Existing incorrect answer: {first_negative}

Return format:

```
{  
  "negatives": ["...", "..."]  
}
```

Prompt 3: Prompt for candidate regeneration.

Generate exactly one additional incorrect answer for this question.

```

Important constraints:
- You are NOT given the ground-truth answer.
- Use only the question and the existing incorrect answer.
- The new answer must preserve the same structural style,
  ↪ granularity, and answer format
  as the existing incorrect answer.
- The new answer must be different from all banned answers.
- It must be plausible and hard, not generic or obviously easy.
- Do not explain.

Question: {question}
Existing incorrect answer: {first_negative}
Banned answers: {banned_answers}

Return format:
{
  "negative": "...

```

Prompt 4: Negative validation.

```

You are validating a hard negative answer for a visual QA benchmark.

A GOOD hard negative must satisfy ALL of the following:
1. It is clearly different in factual content from the ground-truth
  ↪ answer.
2. It preserves the same answer format/structure as the ground-truth
  ↪ answer.
3. It preserves similar granularity/detail level as the ground-truth
  ↪ answer.
4. It is plausible for the scene/question context.
5. It is not a trivial or obviously easy distractor.
6. It should be confusable with the ground-truth by structure/style,
  ↪ but not by factual content.

Question: {question}
Ground-truth answer: {gt_answer}
Candidate negative: {candidate}

Return ONLY one JSON object:
{
  "valid": true,
  "content_changed": true,
  "structure_preserved": true,
  "granularity_preserved": true,
  "plausible": true,
  "too_easy": false,
  "reason": "brief explanation"
}

```

A.2 Evidence Moment Annotation

A central requirement of our streaming evaluation protocol is to know when the visual information needed to answer a question can be observed in the video. We refer to this temporal support as the *evidence moment*. Formally, for each question-answer pair, the evidence moment identifies the timestamp or temporal interval in which the answer can be grounded in the visual stream. This annotation is used to align each question with the relevant observation window and to define subsequent recall settings.

Our evidence-moment construction is dataset-dependent but follows a unified principle: we preserve the temporal grounding provided by each source dataset and convert it into a common representation.

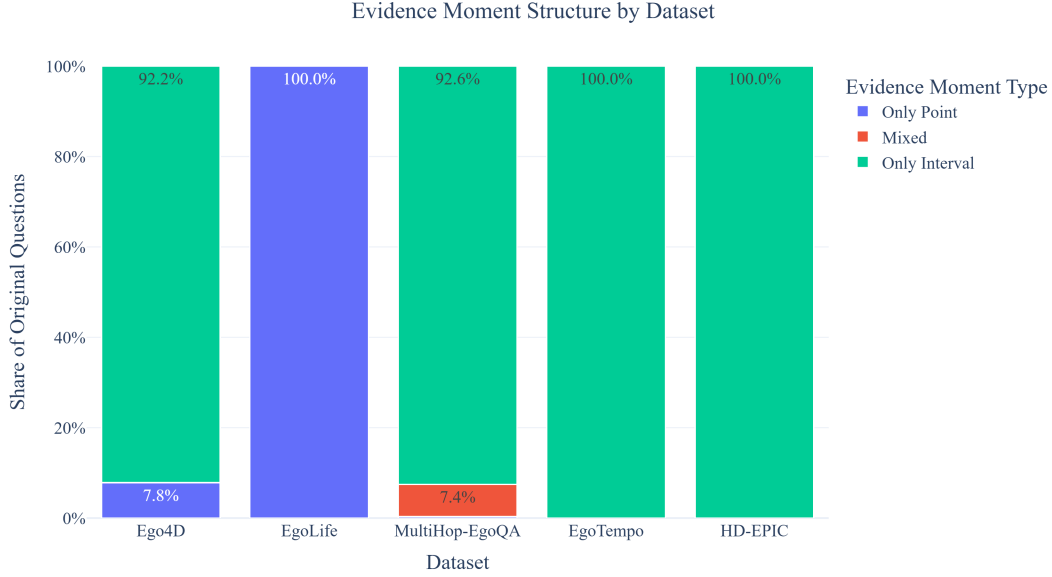


Figure 6: **Evidence moment structure by dataset.** We report the distribution of original questions whose evidence is represented as point-level timestamps, interval-level segments, or mixed evidence.

This avoids introducing unnecessary additional assumptions and keeps the benchmark faithful to the temporal structure of the original annotations.

For **Ego4D Episodic Memory VQA** [3], we use the evidence annotations already provided by the dataset. These annotations identify the temporal support associated with the answer and are directly converted into our unified evidence-moment format.

For **EgoLife** [47], the original temporal grounding is provided as point-level evidence rather than interval-level evidence. We preserve these point annotations as evidence timestamps. In our unified representation, such evidence points serve as temporal anchors at which the answer becomes observable in the stream.

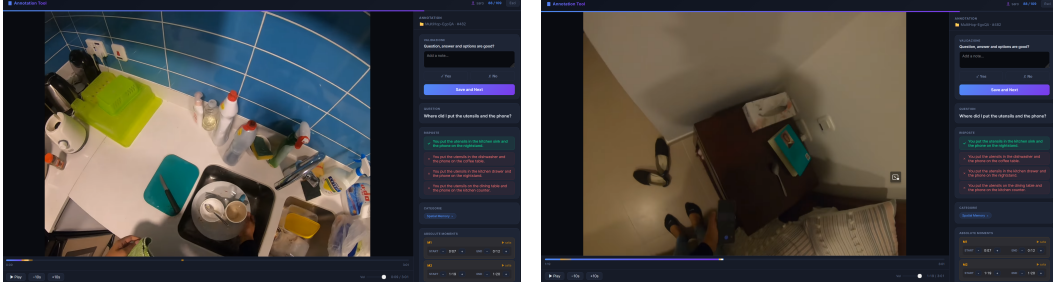
For **Multi-Hop EgoQA** [6], we use the evidence moments provided by the dataset. Since Multi-Hop EgoQA is designed around compositional questions, a single question may require multiple temporally separated pieces of evidence. We therefore retain all annotated supporting moments and represent them as a set of evidence intervals, $\mathcal{M} = \{M_1, \dots, M_n\}$, where each M_i corresponds to one required visual support segment.

For **HD-EPIC** [32], we use the temporally grounded evidence already available in the source annotations. These moments identify the action or event segment from which the answer can be inferred.

For **EgoTempo** [34], the provided clips are trimmed around the action queried by the question. Since each clip is constructed to contain the queried action by design, we treat the full provided clip as the evidence moment. Thus, for an EgoTempo sample with clip duration $[t_s, t_e]$, the evidence moment is defined as $M = [t_s, t_e]$.

Figure 6 summarizes the resulting evidence-moment structure across datasets, highlighting the diversity of the source annotations: EgoLife [47] is entirely point-based, EgoTempo [34] and HD-EPIC [32] are entirely interval-based, while Ego4D Episodic Memory VQA [3] and Multi-Hop EgoQA [6] contain a mixture of evidence structures.

This process yields a consistent temporal grounding format across all five source datasets while respecting their original annotation schemes. In particular, interval annotations are preserved when available, point-level annotations are retained as temporal anchors, multi-hop annotations are represented as sets of evidence moments, and action-centered trimmed clips are treated as full-clip evidence. We paired all 2,634 examples from our hard negative generation step (See Appendix A.1).



(a) Validation view showing the video, the question, answer options, and evidence moments.

(b) Example of the same validation item at a different point in the evidence window.

Figure 7: Human validation interface used to review candidate QA samples. Validators checked whether the question, correct answer, distractors, and temporal evidence were coherent with the video.

A.3 Human Validation

To improve the quality of the benchmark, we performed an additional human validation stage after the automatic and manual curation steps. Validators were asked to inspect each candidate sample through a web-based interface showing the video segment, the question, the correct answer, the distractor answers, and the annotated evidence moments. For each item, validators selected whether the question, answer, and options were valid, while also checking that the relevant action or visual evidence occurred within the annotated evidence moments. Validators could also leave a free-text note explaining the reason for rejection or suggesting a correction.

Figure 7 shows the validation interface. The task was intentionally simple: validators decided whether each sample should be kept or discarded, while viewing the relevant temporal evidence and all answer options. This setup collected both binary quality labels and qualitative feedback on ambiguous questions, incorrect answers, weak distractors, missing visual evidence, or samples that did not clearly test memory. To ensure consistency, annotators received written instructions and example validations before starting. All reviewers participated voluntarily in the human validation process and received no compensation.

Overall, **24 validators** contributed to the validation process, leading to the removal of approximately **12%** of the reviewed annotations, leading our benchmark to 2,335 samples. The free-text notes were useful for identifying recurring quality issues. Discarded samples typically fell into a few common cases: the correct answer was not clearly visible in the video, the question and answers were not well aligned, multiple answer options could be considered correct, distractors were too easy or unrealistic, the question could not be evaluated at multiple recall times, or the question did not actually test memory. Samples marked as invalid were removed from the benchmark.

Figure 8 shows four examples of discarded items. These examples illustrate different failure modes identified during validation, including questions that were not suitable for multiple recall times, questions asking for future predictions rather than memory recall, ambiguous answer choices, and visually unsupported or weakly grounded annotations.

In addition to filtering invalid annotations, human validation confirmed that the retained samples covered a diverse set of memory demands and answer formats. Figure 9 reports representative valid QA pairs from the different source datasets. Each example includes the primary memory label, the correct answer, and the negative options used in the multiple-choice setting.

Invalid recall formulation Dataset: EgoLife Question: <i>When was the meat bought?</i> Correct answer: <i>Yesterday afternoon</i> Distractors: <i>This afternoon; This morning; Yesterday morning</i> Reason for removal: The answer depends on the temporal reference point from which the question is asked, so it cannot be evaluated at multiple recall times.	Does not test memory Dataset: EgoTempo Question: <i>What will the person likely do next after grabbing the envelope?</i> Correct answer: <i>They will likely exit the car.</i> Distractors: <i>They will likely start the engine; They will likely put on their seatbelt; They will likely check the rearview mirror.</i> Reason for removal: The question asks for a likely future action rather than requiring recall of information already observed in the video.
Ambiguous answer options Dataset: MultiHop-EgoQA Question: <i>What did I do with the booklet after picking it up from the stand?</i> Correct answer: <i>After picking the booklet from the stand and carrying it to the seat, you read the booklet.</i> Distractors: <i>After picking the booklet from the stand and carrying it to the seat, you flipped through the booklet; After picking the booklet from the stand and carrying it to the seat, you tucked the booklet into your bag; After picking the booklet from the stand and carrying it to the seat, you briefly examined the cover of the booklet.</i> Reason for removal: The distinction between reading and flipping through the booklet is too subtle to define a reliable single correct answer.	Question-answer mismatch Dataset: HD-EPIC Question: <i>Why did the person open the packaging?</i> Correct answer: <i>By pulling the plastic in opposite direction using both hands.</i> Distractors: <i>By unrolling the bag opening; By unfolding the edges of the packaging holding the cube with the right hand and unfolding with the left hand; By undoing the knot on the top of the packaging using both hands.</i> Reason for removal: The question asks why the action was performed, but the answer options describe how the packaging was opened.

Figure 8: Examples of discarded candidate QA samples. Human validation helped identify samples that were not suitable for the streaming memory protocol, including questions with unstable temporal references, future-prediction questions, ambiguous answer choices, and question-answer mismatches.

A.4 Question Categorization Protocol and Labeling Guidelines

To ensure the high-quality assignment of memory categories across EGOSTREAM, we utilized an iterative Human-in-the-Loop (HITL) pipeline. Crucially, the semantic categorization of each question was performed using only the text of the question and its corresponding answer. This ensured that the taxonomy remained strictly an evaluation of the cognitive demand posed by the query itself, completely independent of the visual evidence. This section details the workflow, the explicit decision rules for resolving overlapping categories, and the guidelines provided to both the LLM and human annotators.

A.4.1 Iterative Human-in-the-Loop (HITL) Workflow

Because episodic queries frequently exhibit overlapping cognitive demands, we assigned each question exactly one *primary* label (representing the core cognitive target) and zero or more *secondary*

Ego4D PRIMARYLABEL: SPAT Q: <i>Where did I put a pair of scissors?</i> Correct: On the table. Negatives: Left of the stove; on the floor; in the drawer.	Ego4D PRIMARYLABEL: DETL Q: <i>What color was the spanner handle?</i> Correct: Black and red. Negatives: Black and yellow; black and white; blue and red.
EgoLife PRIMARYLABEL: PROS Q: <i>What are we planning to eat for lunch today?</i> Correct: Dumplings. Negatives: Hot Pot; Toast; Pizza.	EgoLife PRIMARYLABEL: SOC Q: <i>What does Alice not like to eat?</i> Correct: Steamed Bun. Negatives: Pizza; Rice; Noodles.
EgoTempo PRIMARYLABEL: DETL Q: <i>What does the person use to stir the ham?</i> Correct: Chopsticks. Negatives: A metal spatula; a wooden spoon; a large serving fork.	EgoTempo PRIMARYLABEL: EVNT Q: <i>What object does the person move from the stovetop to the sink?</i> Correct: A frying pan. Negatives: A stainless steel pot; a glass lid; a small saucepan.
MultiHop-EgoQA PRIMARYLABEL: SPAT Q: <i>Where did I put the utensils and the phone?</i> Correct: You put the utensils in the kitchen sink and the phone on the nightstand. Negatives: You put the utensils in the dishwasher and the phone on the coffee table; you put the utensils in the kitchen drawer and the phone on the nightstand; you put the utensils on the dining table and the phone on the kitchen counter.	MultiHop-EgoQA PRIMARYLABEL: SPAT Q: <i>Where did I drop the garment and the yarn during the video?</i> Correct: You dropped the garment on your legs and the yarn on a table. Negatives: You dropped the garment on the chair and the yarn on the floor; you dropped the garment on the bed and the yarn on the table; you dropped the garment on the couch and the yarn on the shelf.
HD-EPIC PRIMARYLABEL: CAUS Q: <i>What was the reason the person shook the bag?</i> Correct: To remove the pieces of onion skin into the food bin. Negatives: So that all of the coffee beans come out and go into the grinder; so that all of the raisins come out; to mix in the salt; to open the bag fully.	HD-EPIC PRIMARYLABEL: CAUS Q: <i>What was the reason the person tapped the phone keyboard?</i> Correct: To make sure the nutritional value of the spice is correctly recorded. Negatives: To find the couscous measurements in the phone app; to record measurements of salad; to drop food residue attached to the spatula on the frying pan; to drop the remaining soup from the ladle into the container.

Figure 9: Examples of valid QA pairs included in the benchmark. Each card reports the source dataset, the primary memory label, the question, the correct answer, and the negative options. The examples illustrate the diversity of memory demands covered by the benchmark.

tags (capturing additional contextual dimensions). The annotation process followed an iterative cycle to scale efficiently while guaranteeing human-level accuracy:

1. **Initial LLM Prediction (round 1):** A Large Language Model (Gemini 2.5 Pro) was prompted to assign initial memory categories based on early rule drafts. The used prompt is reported in Prompt 1.
2. **Manual Correction (round 1):** Human expert annotators then manually reviewed and corrected a large subset of these predictions (847 questions). During this phase, critical ambiguities were identified and resolved (e.g., distinguishing between a question asking for a spatial location vs. a question using a spatial phrase merely to identify an object). If some questions were considered low quality or malformed, annotators had the ability to discard them. Figure 10 shows a screenshot of the interface used for this refinement.
3. **Automated Tagging with In-Context Learning (round 2):** The manually corrected, high-quality seed set was then used to craft a refined prompt defining precise cascade rules and include few-shot, in-context examples to make automated tagging more accurate. This guided a subsequent round of LLM labeling (Gemini 2.5 Pro), significantly reducing the

ann_1861

How did the person pick up the bowl of coconut milk?

Original: EVNT Working: EVNT Human: EVNT Source: HD-EPIC

Secondary: none

Latest AI: EVNT (0.95) The question asks for the procedural method of picking up an object.

✓ Correct

using left hand and removing the fork used to stir it using right hand.

✗ Wrong

using both hands holding the bowl from the bowl rim.

✗ Wrong

by holding both sides using the oven gloves.

✗ Wrong

using the right hand and lift the large white bowl up.

✗ Wrong

using both hands from the kitchen top above the dishwasher.

SPAT
Spatial Memory

TEMP
Temporal Memory

DETL
Fine-Grained
Details

EVNT
Event Memory

SOC
Social Memory

CAUS
Causal Reasoning

PROS
Prospective
Memory

PROC
Task Schema &
Procedural

DISCARD
Discard

Review notes

Reasoning, ambiguity notes, rule references...

Accept (A)

Save Label Change (S)

Flag Ambiguous (F)

Skip (K)

Next (N)

← Undo (U)

Figure 10: Interface used to refine category annotations for questions. The interface allowed to accept the automated labels, or change them, including a textual description.

initial error rate. In this stage, all questions are assigned a revised category, but original human labels are retained. The prompt is reported in Prompt 2.

4. **Full Human Verification (round 2):** At this stage, human annotators used the interface introduced in Figure 10 to 1) revise old questions where human and AI disagreed and 2) revise new questions with no human labels. At this stage, labeling is faster, as annotators just have to confirm automated labels in most cases.

Figure 11 shows the evolution of AI-generated and human labels obtained with the described process. The process starts with 2,335 questions. By the end of the process, 85 questions were discarded, leaving with a set of 2,250 questions.

A.4.2 Primary Labeling: The Priority Cascade

The process described above aims to obtain primary labels isolating the single most dominant cognitive demand of a question. To resolve overlaps consistently during both LLM prompting and human review, we formalized a strict deterministic priority cascade. Annotators evaluated the question text against the following list in order, stopping at the first match:

1. **DISCARD (Malformed/Generic):** Exits the cascade if the question is out-of-scope or asks for generic procedural knowledge (e.g., “How do I usually buy tickets?”) rather than episodic recall.
2. **CAUS (Causal):** Applies if the question explicitly asks for a reason or explanation (“Why...?”).
3. **PROS (Prospective):** Applies if the target is a future-oriented plan or intended next action, even if objects or people are involved.

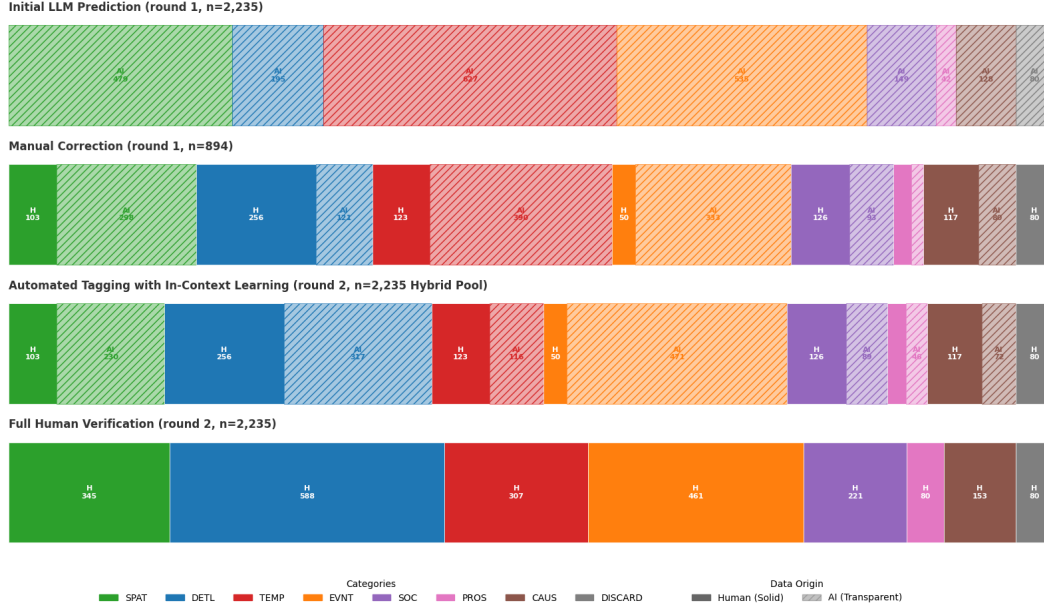


Figure 11: Evolution of AI-generated and human labels in the HITL process.

4. SOC (**Social**): Applies if the expected answer is a person or group (“Who...?”). Social targets supersede temporal or spatial context.
5. TEMP (**Temporal**): Applies if the core target is time, ordering, or a sequence (“When...?”, “In what order...?”). Time-anchored states (“before I picked it”) default to TEMP unless the query explicitly asks for a person.
6. SPAT (**Spatial**): Applies when the object is known, and the target answer is a physical location (“Where is X?”).
7. DETL (**Detail**): Applies when the question asks for fine-grained content (identity, color, count). *Crucial Tie-Break*: If a question uses a location purely to index an object (e.g., “What did I put in the microwave?”), the label is DETL, not SPAT, because the expected answer is an object identity.
8. EVNT (**Event**): Serves as the residual category for questions probing action gist, state changes, or execution (“How did I do X?”, “Did Y occur?”) that do not cleanly fit the more specific categories above.

A.4.3 Secondary Labeling

To capture the multidimensional nature of episodic memory, zero or more *secondary* tags were assigned after the primary label was fixed. These were automatically assigned with an LLM (Gemini 2.5 Pro) using Prompt 3 and then manually refined. Unlike primary labels, secondary tags do not use a priority cascade or tie-breaking rules. Any cognitive dimension that is explicitly required by the question text—other than the primary label—is tagged.

The following examples illustrate how primary and secondary labels are assigned across different source datasets:

- **Ego4D** – “Where did I put a pair of scissors?” → **Primary**: SPAT (target is the final location of the scissors). **Secondary**: EVNT (location is defined relative to a past placement action).
- **Ego4D** – “What did I pour in the dish?” → **Primary**: DETL (fine-grained identity of the poured content). **Secondary**: SPAT (tied to a specific container), EVNT (embedded in a pouring event).
- **Ego4D** – “how many planks were on the floor?” → **Primary**: DETL (object count). **Secondary**: SPAT (planks are defined by their location “on the floor”).

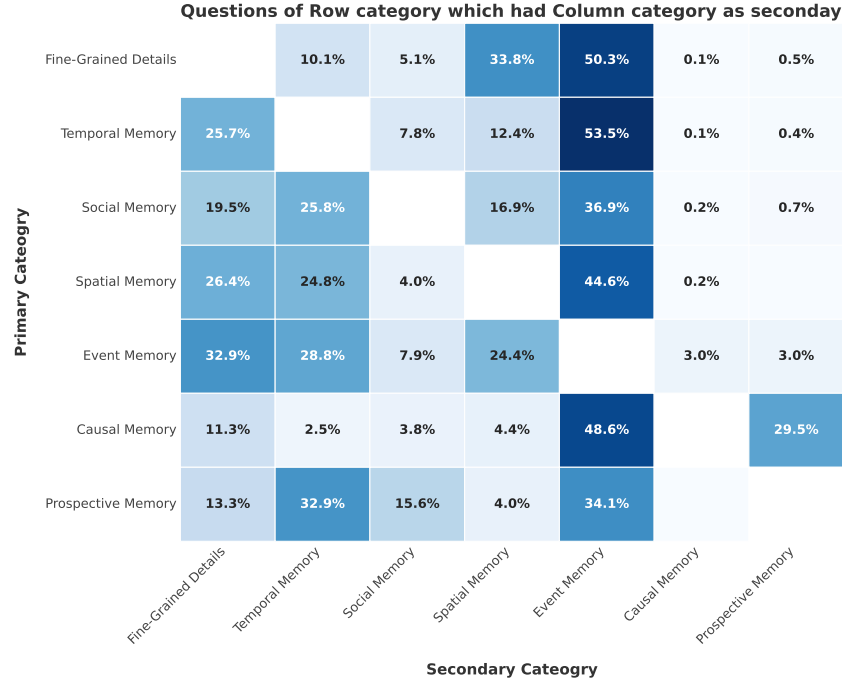


Figure 12: Co-occurrences between primary and secondary labels. Each cell shows the percentage of questions which are primarily marked with the category in the rows, but are also tagged with the category on the column.

- **Ego4D** – “*what did I put in the wood plank?*” → **Primary:** DETL (object inserted into the plank). **Secondary:** EVNT, SPAT (object detail queried within a spatially anchored manipulation event).
- **EgoLife** – “*Who was with me when I visited the apartment?*” → **Primary:** SOC (person identity). **Secondary:** TEMP (anchored to a specific visit), EVNT (bound to the visiting event).
- **EgoTempo** – “*What did I do after putting the chopped dill in the food?*” → **Primary:** TEMP (ordering of actions). **Secondary:** EVNT (reasoning over the surrounding procedure), DETL (implicit reference to the dill and food state).
- **Multi-Hop EgoQA** – “*Why didn’t we buy the chestnut kernels?*” → **Primary:** CAUS (reason for a decision). **Secondary:** EVNT (shopping episode), DETL (specific item).
- **HD-EPIC** – “*What should I buy to decorate my room?*” → **Primary:** PROS (future-oriented plan). **Secondary:** DETL (specific items), SPAT (target space: the room).

By decoupling the strict primary classification from the inclusive secondary tags, our annotation protocol supports both clean, mutually exclusive category evaluation and richer, multi-label diagnostic analysis. Figure 12 shows the co-occurrences between primary and secondary labels. As can be noted, most questions do not strictly belong to a single category, with “Event Memory” being the most overlapped with other categories as a secondary tag. This is expected, as most properties of episodic memory refer to specific events.

Prompt 5: Initial prompt for assigning primary question category.

```
# EgoStream Memory Classification Prompt (v1)

## Task

Given a list of egocentric VideoQA questions, assign exactly one
↪ cognitive category to each question from the taxonomy below.
```


Table 6: Compact qualitative examples across memory dimensions. Examples illustrate how questions are assigned primary and secondary memory labels, while also covering different Answer Validity Window (AVW) ranges.

Frame	Question and correct answer	Labels / AVW	Dataset
	Q: Where is the roll of blue tape at the beginning of the video? A: The roll of blue tape is on the white wooden board.	Primary: SPAT Secondary: DETL TEMP AVW: Instant	EgoTempo
	Q: Who closed the green specimen tray and the white bottle during the video? A: You closed the specimen tray and the white bottle.	Primary: SOC Secondary: EVNT DETL AVW: 1s	MultiHop
	Q: Who was with me when I was machining the engine body? A: A man with blue pants.	Primary: SOC Secondary: TEMP EVNT AVW: 10s	Ego4D
	Q: How did the person put the spoon? A: By slotting the spoon handle into the pocket of an apron.	Primary: DETL Secondary: EVNT AVW: 42s	HD-EPIC
	Q: Where was the steel spoon before I put the card game instructions in the paper bag? A: Red cup.	Primary: SPAT Secondary: TEMP EVNT DETL AVW: 2.5m	Ego4D
	Q: Where was I the last time I pushed a cart with a friend? A: Hema Supermarket.	Primary: SPAT Secondary: TEMP EVNT SOC AVW: 13.8h	EgoLife

Label Legend: **SPAT** Spatial, **DETL** Detail, **TEMP** Temporal, **EVNT** Event, **SOC** Social, **CAUS** Causal, **PROS** Prospective.

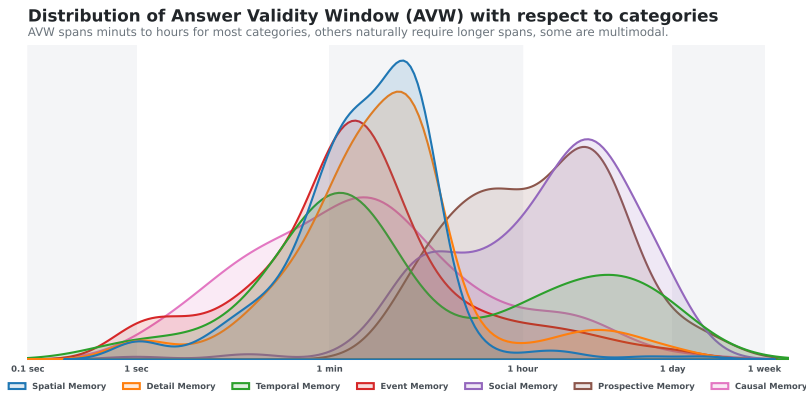


Figure 13: Distribution of Answer Validity Window across Categories.

Always label the category that reflects the question's **primary**
↳ **cognitive probe**: the main memory system or reasoning demand the
↳ question is designed to stress, not just superficial wording.

General principles

- **Single label**: Each question must receive one and only one
↳ category.
- **Primary probe**: When multiple aspects are present (e.g., time
↳ and space), choose the category that captures the *hardest* or
↳ most central cognitive requirement.
- **Ignore surface form**: Do not rely only on wh-words; e.g., not
↳ every "Where" is SPAT, and not every "How many" is TEMP. Use the
↳ rules and examples below.
- **Retrospective vs prospective**: Distinguish questions about
↳ **past events** (retrospective) from questions about
↳ **remembering to perform an intended future action**
↳ (prospective).

Taxonomy Definitions

1. Spatial Memory (SPAT)

- **Focus**: Current or last known location of objects or entities in
↳ space.
- **Primary probe**: Accessing the agent's spatial map--"Where is X
↳ now?" or "Where did we leave X (in the last valid state)?"
- **Typical question forms**:
 - "Where is the screwdriver now?"
 - "On which side of the sink is the mug?"
- **Inclusion**:
 - Questions whose answer depends mainly on knowing *where* something
↳ is in the environment at the current or last valid time.
- **Exclusion / tie-breaker**:
 - If the question explicitly refers to a **relative time anchor**
↳ ("before", "after", "earlier"), classify as **TEMP**, because
↳ the main difficulty is retrieving a prior world state given its
↳ time index.

2. Temporal Memory (TEMP)

- **Focus**: Sequence, relative order, timing, duration, and
↳ frequency of **events**.
- **Primary probe**: Navigating the temporal stream--"When",
↳ "Before/After", "In what order", "How long", "How many times".
- **Typical question forms**:
 - "In what order did I open the fridge and the drawer?"
 - "How long was the stove on before I turned it off?"
 - "How many times did I go to the kitchen?" (count of **event**
↳ occurrences).
 - "Where was the spanner before I carried it?" (spatial phrase but
↳ temporal pivot "before").
- **Inclusion**:
 - Questions about **order**, **relative time**, **event**
↳ frequency, or **past locations indexed by time**.
- **Exclusion / tie-breaker**:
 - Counts of **objects** ("How many eggs did I crack?") → **DETL**,
↳ because this targets high-fidelity numerical detail rather than
↳ temporal structure.

- Questions that only ask "When will I do X in the future?"
 - ↳ *without* checking whether an intention was remembered/executed
 - ↳ are usually better treated as CAUS/EVNT, not PROS.

3. Fine-Grained Details (DETL)

- **Focus**: High-fidelity perceptual or specific factual details:
 - ↳ visual features, object identity, and counts of objects.
- **Primary probe**: Can the model recall precise attributes rather than just the gist (color, exact item, count)?
- **Typical question forms**:
 - "What color was the mug?"
 - "How many eggs did I crack?" (count of eggs as objects).
 - "What did I pour into the dish?" (when the answer is a specific substance like "detergent").
- **Inclusion**:
 - Questions whose answer is a **specific attribute or identity** (color, shape, brand, material, exact tool type) or **object count**, even if events span time.
- **Exclusion / tie-breaker**:
 - If the question is about the **gist of an action** ("What did I do after entering the kitchen?") rather than attributes of objects, use **EVNT**.
 - If the question asks "Why" something happened, even if details are involved, use **CAUS**.

4. Event Memory (EVNT)

- **Focus**: Retrospective recall of the **gist** of actions or episodes ("what happened"), without emphasizing fine detail, spatial layout, temporal ordering, or causal explanation.
- **Primary probe**: Can the model recall and summarize what the user did or what occurred at a certain time?
- **Typical question forms**:
 - "What did I do after entering the kitchen?"
 - "What happened when I went to the balcony?"
- **Inclusion**:
 - Questions whose main goal is to retrieve **what** action(s) occurred, at a gist/event level.
- **Exclusion / tie-breaker**: If a question primarily targets:
 - **Spatial location** → SPAT
 - **Temporal order/duration/frequency** → TEMP
 - **Perceptual details** → DETL
 - **Social identity or interaction** → SOC
 - **Causal explanation/motivation** → CAUS
 - **Remembering to execute a future intention** → PROS

then use the more specific category instead of EVNT.

5. Social Memory (SOC)

- **Focus**: People, identities, and social interactions.
- **Primary probe**: Who was involved, who did what, what someone said or did.
- **Typical question forms**:
 - "Who was talking to me in the hallway?"
 - "What did Anna say before we left?"
- **Inclusion**:

- Questions where recognizing or recalling **people** or their
 ↳ **utterances/actions** is central (faces, names, roles,
 ↳ specific interlocutors).
- **Exclusion / tie-breaker**:
 - If the main point is **why** a person did something ("Why did Anna leave?"), use **CAUS**.
 - If the person is incidental and the main cognitive demand is
 ↳ spatial/temporal/detail, use SPAT/TEMP/DETL.

6. Causal Reasoning (CAUS)

- **Focus**: Inferring causes, motivations, or consequences of
 ↳ actions or states.
- **Primary probe**: Explaining **why** something happened or what it
 ↳ leads to, not just recalling that it happened.
- **Typical question forms**:
 - "Why did I go to the kitchen?" (motivation for an action).
 - "Why did the pot boil over?" (linking previous actions to an
 ↳ outcome).
- **Inclusion**:
 - "Why...", "What caused...", "What was the reason...", "What was
 ↳ the consequence of..."
 - Questions whose correct answer requires integrating multiple
 ↳ observations to infer a cause or intent.
- **Exclusion / tie-breaker**:
 - Questions only asking **what happened** or **who did what**
 ↳ without a causal explanation → **EVNT** or **SOC**.

7. Prospective Memory (PROS)

- **Focus**: Remembering to perform **planned actions** in the
 ↳ future--linking previously formed intentions to later execution
 ↳ at the right time or cue.
- **Primary probe**: Did the agent remember and carry out a prior
 ↳ intention?
- **Typical question forms**:
 - "Did I remember to lock the door when I left, as I planned
 ↳ earlier?"
 - "Did I take my medicine at 8 pm like I intended?"
 - "What was I planning to do after finishing this call?" (if the
 ↳ intention was set earlier).
- **Inclusion**:
 - Questions that explicitly reference a **prior intention** and a
 ↳ **later check** on whether it was remembered/executed.
- **Exclusion / tie-breaker**:
 - Questions that merely describe future plans without checking
 ↳ remembering ("What am I going to do later?" asked at planning
 ↳ time only) are better assigned to EVNT/CAUS depending on
 ↳ content.
 - Questions about past events with no future-intention component
 ↳ belong to TEMP/EVNT/others, not PROS.

Corner Cases & Tie-Breaker Table

Question	Possible categories	Decision & Rationale
:---	:---	:---
"What did I pour in the dish?"	EVNT vs DETL	DETL : The ↳ answer is the specific identity of the substance (e.g., ↳ detergent), a high-fidelity object detail rather than just the ↳ action "I poured something."

```

| "Where was the spanner before I carried it?" | SPAT vs TEMP |
↳ **TEMP**: The key challenge is retrieving a **past state**
↳ indexed by "before", i.e., temporal navigation, even though the
↳ answer is a location. |
| "How many eggs did I crack?" | TEMP vs DETL | **DETL**: This is a
↳ **count of objects**, not a count of event occurrences or visits,
↳ so we treat it as fine-grained detail. |
| "How many times did I go to the kitchen?" | TEMP vs EVNT |
↳ **TEMP**: This is a **frequency of events**, which directly
↳ probes temporal/event counting. |
| "Why did I go to the kitchen?" | EVNT vs CAUS | **CAUS**: The
↳ question asks for the **motivation or reason** for the action,
↳ not just what happened. |
| "Did I remember to turn off the stove after deciding to cook pasta
↳ later?" | TEMP vs PROS | **PROS**: This is explicitly about
↳ whether a **previous intention** was remembered and executed at
↳ the appropriate later time. |

***

## Input Format

You will receive as input a json file `input_questions.json`.

***

## Output Format

Return a JSON array `input_questions_automated_round1.json` following
↳ the format of `input_questions.json`. Keep only one category per
↳ action decided according to the rules above.

Add rationales for your choices and keep them short but specific (1-2
↳ sentences), linking the question to the rules above.

```

Prompt 6: Refined prompt for assigning primary question category.

```

# EgoStream Memory Classification Prompt (v2)

You are given **questions about an egocentric episode** (first-person
↳ video).
Your task is to assign **exactly one primary category** to each
↳ question from this set:

- `DETL` - Fine-grained details
- `SPAT` - Spatial memory
- `TEMP` - Temporal memory
- `EVNT` - Event memory
- `SOC` - Social memory
- `PROS` - Prospective memory
- `CAUS` - Causal memory
- `DISCARD` - Out-of-scope / malformed

Follow the **decision cascade** below in order. At each step, if a
↳ rule matches, assign that label and stop.

---

## 1. DISCARD filter (early exit)

Assign `DISCARD` if:

- The question is malformed, incoherent, or clearly unanswerable from
↳ the episode.

```

- The question is **generic procedural knowledge** rather than
 ↳ episode-specific, e.g.:
 - "How do I usually buy plane tickets?"
 - "What is the best way to scan my house for things I forgot?"

If in doubt, prefer a real category rather than DISCARD; use DISCARD
 ↳ only when the question truly does not fit any memory category or
 ↳ is nonsensical.

****If DISCARD applies, stop.****

2. CAUS - Causal memory

Assign `CAUS` if the **primary purpose** of the question is to ask
 ↳ for a **reason or explanation**, typically introduced by "why" or
 ↳ "for what reason", and the expected answer is a **cause**, not a
 ↳ procedure.

Examples:

- "Why didn't we buy chestnut kernels?"
- "Why did we put the durian back?"

If the question is **"how did I do X?"** and the answer is an
 ↳ **action sequence** (not a reason), this is **EVNT**, not CAUS.

****If CAUS applies, stop.****

3. PROS - Prospective memory

Assign `PROS` when the question asks about **plans, intentions, or**
 ↳ **future actions** (things that are supposed to happen or be done).

Typical patterns:

- "What are we planning to eat for lunch?"
- "What are we going to do tomorrow?"
- "What does Lucia need to buy to decorate her room?"
- "Who plans to grow flowers?" (future-oriented social plan → still
 ↳ PROS)

Prospective trumps other dimensions (including SOC) when the core
 ↳ target is an intended future action or plan.

****If PROS applies, stop.****

4. SOC - Social memory

Assign `SOC` when the **answer is one or more people** (identity of a
 ↳ person or group), or a stable property of people in the episode.

Use SOC whenever **"who"** is the main query, even if there are
 ↳ temporal or contextual cues.

Examples:

- "Who did I talk to in the workshop?"
- "Who did I talk to in the room?"
- "Who did I lose track of at the supermarket on day 1?"
- "Who was next to me the last time I was taste-testing something?"

```

- "Who usually picks up the takeout?"
- "Who usually frequents the vegetable market?"

Even if the question mentions time ("last time...", "on day 1...") or
↳ place, if the target is the person, use `SOC`.

If SOC applies, stop.

---

## 5. TEMP - Temporal memory

Assign `TEMP` when the core target is time or order of
↳ events: when something happened, in what sequence, or at which
↳ occurrence ("last time", "first time", etc.). The expected answer
↳ is a time, temporal position, or ordering.

Typical patterns:

- Explicit when/last time/first time:
  - "When was the meat bought?"
  - "When was the last time eggs were bought?"
  - "When did we finish washing the dishes?"

- Order / sequence:
  - "In what order did I open the dishwasher and the container?"

- Location in a temporal chain, where time is primary:
  - "Where was the knife before I picked it?" → `TEMP`
    (You are being tested on recalling an earlier state indexed by
    ↳ time "before", not just "where is it now".)

Use this tie-breaker:

> If the question's main point is "at what time / in what order /
↳ which occurrence", label `TEMP`, even if the answer mentions a
↳ location or object.

Exceptions (override TEMP):

- If the question is clearly about who (person) and "last time"
↳ is just context, use `SOC`.
- If "before/after" is only a weak cue and the answer space is
↳ purely locations, and the annotator's intent is clearly
↳ spatial, `SPAT` can trump TEMP (see SPAT below).

If TEMP applies, stop.

---

## 6. SPAT - Spatial memory

Assign `SPAT` when the target of the question is a location or
↳ spatial configuration. The answer should be "where"
↳ something is or was, or a surface/region relation ("on", "in",
↳ "next to", etc.), not which object it is.

Key pattern:

> The object is given, and the question asks where it is/was.

Examples:

- "Where is my dreamers yora?"
- "Where is the stool?"
- "Where did I put the shoe?"

```

- "Where did I put the cabbage?"
- "Where did I put the chopsticks?"
- "Where did I put the books?"
- "Where was my phone?"
- "In what location did I see the orange couch?"
- "In what location did I see the toolbox?"
- "Where did I carry the carton to?"
- "Where was the AD calcium milk taken from?"

For "before/after" spatial questions:

- "Where was the candle jar before I operated my phone?" → `SPAT`
(Here the practical focus is the location, time just anchors which
↪ state.)

Tie-break rule vs DETL:

- If the **answer** is a location** (shelf, table, room, "next to the
↪ sink", etc.), choose `SPAT`.
- If the **answer** is an object identity, color, or count**, even if
↪ the question contains "in/on/at [location]", choose `DETL` (see
↪ next section).

If SPAT applies, stop.

7. DETL - Fine-grained detail memory

Assign `DETL` when the question asks for **fine-grained content**:

- Identity of an object
- Color
- Number / count
- Type / category / brand
- Text written on an object
- Specific item used/consumed/etc.

Crucially, **DETL** includes questions where a location/time is used
↪ only as a cue to select the object**, but the real target is the
↪ object or its properties.

7.1 Typical DETL patterns

Object identity:

- "What did I pour in the dish?"
- "What did I pour in the cup?"
- "What did I put in the saucepan?"
- "What did I put in the microwave?"
- "What did I put in the bucket?" (identity of what was put in)
- "What vegetable did I slice?"
- "What ingredient did I pick?"
- "What tool did I remove from the drawer?"
- "What cable did I connect to the circular saw?"

Color / appearance:

- "What color was the spanner handle?"
- "What color is the towel?"
- "What color is the air fryer?"
- "What color was the machine on the floor?"
- "What color was the jerrycan on the shelf?"
- "What color was the wallpaper I picked from the box?"
- "What color is the washing machine that I touched?"

```

**Number / count:**

- "How many oven trays did I take from the trolley?"
- "How many pastries did I put on the tray?"
- "How many planks were on the floor?"
- "How many blue bottles did I see?"
- "How many cables did I pick?"
- "How many spoons were in the cooking pot?"

**Text / label / semantic content:**

- "What words were written on the box close to the wall?"

**Preferences / stable content facts (when answer is not a person):**

- "What kind of sauce do Shure and I usually have with hot pot?"
- "What brand of mineral water do we typically buy?"

### 7.2 Location as index → still DETL

If a question uses location or time to specify which object we
↳ are talking about, but the answer is still the object
↳ identity/color/number**, use `DETL`.

Examples:

- "What color was the cup on the wall?" → DETL
- "What color was the drum on the floor?" → DETL
- "What color is the shopping bag on the floor in the room?" → DETL
- "What color was the cloth on the carton?" → DETL
- "What did I put in the bucket?" → DETL (identity of object)
- "What did I pour into the steel bowl?" → DETL

**Rule of thumb for DETL vs SPAT:**

- If the answer is an object identity, color, or count → DETL.
- If the answer is a location → SPAT.

### 7.3 DETL vs EVNT

Use `DETL` when the question is "which specific object/value?",
↳ even if phrased as "what did I X?":

- "What vegetable did I slice?"
- "What did I use to stir the soup?" (identity of utensil)
- "What ingredient did I pick?"

Use `EVNT` (see next section) when the question is about what
↳ happened in an event or how an action was performed**, not about
↳ which object.

If DETL applies, stop.

---

## 8. EVNT - Event memory

Assign `EVNT` when the question targets the event/gist itself:
↳ what happened, how it unfolded, or whether a particular
↳ action/state occurred. It is not about fine object details, nor
↳ about pure when/where/who.

Typical patterns:

- How an action was done (procedural within the episode):
  - "How did the person open the coffee beans bag?"

```



```

- "How did the person remove the excess water?"
- "How did I open the recipe kit yellow tub?"
- "How did I take the nails from storage?"

- **Gist of a past episode or outcome:**
- "What did we end up eating the last time we ordered takeout?"
  (overall event outcome, not a specific ingredient color/count)
- "What items are still not bought?" (status of a plan over time)

- **Binary / state questions about whether something happened or
↳ how:**
- "Did I leave the door open?"
- "Did I close the fridge door after taking the milk?"

```

Use EVNT as a **residual category** when:

```

- The question is clearly about the action or state change
↳ itself, and
- It does not fit CAUS, PROS, SOC, TEMP, SPAT, or DETL more
↳ specifically.

```

9. Summary of priority order

Always apply categories in this **strict order**; once matched, stop:

```

1. `DISCARD` - malformed / out-of-scope generic
2. `CAUS` - why / reason questions
3. `PROS` - future plans / intended actions
4. `SOC` - answer is person/people
5. `TEMP` - when / order / last time (time answer)
6. `SPAT` - where / spatial relation (location answer)
7. `DETL` - object identity, color, number, type, text, detailed
↳ content
8. `EVNT` - event gist: what happened, how it was done, or whether it
↳ occurred

```

When in doubt:

```

- Ask: What type of thing is the answer?
- Person → SOC
- Time / order → TEMP
- Location → SPAT
- Object identity / color / count / text → DETL
- Reason → CAUS
- Future plan → PROS
- Otherwise → EVNT

```

Prompt 7: Prompt to assign secondary tags to questions.

Secondary Memory Tags for Episodic QA

You are given questions about a specific egocentric episode and a
↳ fixed primary label in `primaryLabel`.

Primary labels come from this set:

```

- `DETL` Fine-grained details
- `SPAT` Spatial memory
- `TEMP` Temporal memory
- `EVNT` Event memory
- `SOC` Social memory
- `PROS` Prospective memory

```

```

- `CAUS` Causal memory
- `DISCARD` Out-of-scope or malformed

## Goal
Assign `secondaryLabels`: zero or more additional labels from the
↪ same set that also apply.

## Hard constraints
1. Never change the primary label.
2. Secondary labels can be empty.
3. Never include the primary label inside `secondaryLabels`.
4. Use only the question text and primary label.
5. If `primaryLabel` is `DISCARD`, `secondaryLabels` must be empty.
6. Return labels as a set (no duplicates).

## Semantics for secondary labeling
- Add `DETL` when fine-grained item identity/count/text/color is
  ↪ truly required.
- Add `SPAT` when location or spatial relation is essential.
- Add `TEMP` when order/time anchors (before/after/last/first/when)
  ↪ are essential.
- Add `EVNT` when whether/how an action happened is a meaningful
  ↪ dimension.
- Add `SOC` when person identity/co-presence/role is meaningful.
- Add `PROS` when intent/plan/likely next action is meaningful.
- Add `CAUS` when reason/explanation is meaningful.
- Do not add `DISCARD` secondarily.

## Input format
You will receive a JSON array where each item has:
- `id`
- `question`
- `primaryLabel`

## Output format
Return only a JSON array with one object per input item:

[
  {
    "id": "ann_1",
    "secondaryLabels": ["TEMP", "SPAT"]
  }
]

No markdown. No extra keys. No prose.

```

A.5 Answer Validity Window and Recall Regimes

A key objective of EGOSTREAM is to evaluate not only whether a model can answer a question, but also how long the required information remains valid and retrievable in a streaming setting. We therefore associate each question-answer pair with an *Answer Validity Window* (AVW), defined as the duration for which the annotated answer remains valid after the supporting evidence has been observed.

Let $\mathcal{M} = \{M_1, \dots, M_n\}$ be the set of evidence moments for a question-answer pair, where each moment is either an interval $M_i = [s_i, e_i]$ or a point timestamp represented as a degenerate interval. We define the evidence completion time as

$$t_{\text{ev}} = \max_i e_i,$$

i.e., the time at which all evidence required to answer the question has appeared. The Answer Validity Window is then

$$\text{AVW} = t_{\text{valid}} - t_{\text{ev}},$$

where t_{valid} is the last timestamp at which the annotated answer is still valid. A query at time t_q is admissible only if

$$t_{\text{ev}} \leq t_q \leq t_{\text{valid}}.$$

The recall delay is

$$\Delta t = t_q - t_{\text{ev}},$$

Since admissible queries must satisfy $t_{\text{ev}} \leq t_q \leq t_{\text{valid}}$, the set of possible recall delays for a given question is

$$0 \leq \Delta t \leq \text{AVW}.$$

Thus, the AVW defines the maximum admissible recall delay: instant recall queries are issued at evidence completion, while the largest possible delayed query occurs at t_{valid} .

The definition of t_{valid} depends on the temporal structure of each source dataset. For **EgoLife** [47], which provides point-level evidence and a query timestamp, we set $t_{\text{valid}} = t_q$ and compute $\text{AVW} = t_q - t_{\text{ev}}$. For **EgoTempo** [34], clips are centered around the queried action and evaluated immediately after it; thus $t_q = t_{\text{ev}}$ and $\text{AVW} = 0$, placing all samples in the instant-recall regime.

For **Ego4D Episodic Memory VQA** [3] and **Multi-Hop EgoQA** [6], we use the temporal evidence provided by the source annotations. In the single-evidence case, t_{ev} is the end of the annotated supporting moment; for Multi-Hop EgoQA, it is the end of the last required evidence segment. Since both datasets are originally designed for offline VideoQA, the answer is assumed to remain valid until the end of the clip. We therefore set t_{valid} to the final timestamp of the clip.

For **HD-EPIC** [32], we derive validity boundaries from the narration stream. We convert narrations into SRT-like timestamped entries, split them into overlapping chunks, and query Gemini 3.1 [38] with the memory question and narration text using Prompt 8. The model returns candidate answerable spans

$$\mathcal{C} = \{M_i = [s_i, e_i]\},$$

together with supporting narration lines, confidence, and a short evidence description. We keep medium- and high-confidence spans, merge adjacent predictions, and remove duplicates. A candidate span M_x is considered reliable if it overlaps the ground-truth evidence interval M_{gt} with

$$\text{IoU}(M_x, M_{\text{gt}}) > 0.5.$$

To ensure that the ground-truth answer remains the only valid answer during inference, we trim the video between the neighboring answerable spans. If $M_{x-1} = [s_{x-1}, e_{x-1}]$ and $M_{x+1} = [s_{x+1}, e_{x+1}]$ are the previous and next candidate spans for the same question, the evaluation subclip is defined as

$$[e_{x-1}, s_{x+1}].$$

We then set t_q to the last second of this subclip and compute $\text{AVW} = t_q - t_{\text{ev}}$ as above. This yields an unambiguous validity-preserving clip while keeping recall delay measured from evidence completion.

Prompt 8: HD-EPIC narration-based validity span grounding

```
You are a temporal grounding engine that works ONLY from narrated
↳ subtitles (SRT-like text).
Your job is to locate candidate time spans where a given memory
↳ question is answerable based on the narration.

You will be given:
- a memory_nlg_question (natural language memory question)
- a narration_srt_chunk (SRT entries with timestamps)

TASK
Identify ALL candidate time spans within the provided
↳ narration_srt_chunk where the question is answerable.
A candidate span is valid if the narration in that interval provides
↳ enough textual evidence to answer the question.
Return multiple spans if the question is answerable multiple times in
↳ the chunk.
If the chunk contains no answerable evidence, return an empty list.
```

```

IMPORTANT RULES
- Use ONLY the narration text provided in narration_srt_chunk. Do NOT
  ↳ assume visual information.
- Do NOT invent events or add details not present in the narration.
- Do NOT use or infer any answer choices.
- A span should be as tight as possible, while still covering the
  ↳ evidence needed to answer the question.
- If evidence is spread across adjacent lines, extend the span to
  ↳ include all required adjacent lines.
- Timestamps MUST come from the SRT entries.
- Use the earliest start time and latest end time of the supporting
  ↳ lines.
- supporting_line_idxes must list the SRT indices used as evidence.
- Confidence:
  - "high" = the narration directly provides enough evidence to
    ↳ answer the question
  - "medium" = the narration likely provides enough evidence, but
    ↳ requires mild inference
  - "low" = the narration is only weakly relevant
- If you are unsure, prefer returning no span.

OUTPUT FORMAT
Return exactly ONE valid JSON object and nothing else.
Use this schema:

{
  "spans": [
    {
      "start_sec": <float>,
      "end_sec": <float>,
      "supporting_line_idxes": [<int>, ...],
      "confidence": "low" | "medium" | "high",
      "evidence": "<short quote or paraphrase from the narration>"
    }
  ]
}

If there are no answerable spans in this chunk, return:

{
  "spans": []
}

INPUT:
{
  "question": "{memory_question}",
  "narration_srt_chunk": "{srt_chunk}"
}

```

A.6 Recall Regimes

Once an Answer Validity Windows are assigned, we derive recall-conditioned evaluations by sampling query times within the admissible validity duration. This ensures that delayed queries measure memory retention rather than answer invalidation. If a model fails at a delayed query time, the annotated answer is still valid by construction, so the failure can be interpreted as degradation under increasing temporal distance.

We define recall regimes using a psychology-inspired, approximately logarithmic temporal scale. Our design follows the classical paradigm of probing memory at multiple delays to measure retention and forgetting, dating back to Ebbinghaus' forgetting-curve studies [13, 30]. It also reflects the distinction between short-lived working-memory processes, typically operating over seconds [1], and longer consolidation processes unfolding over minutes, hours, and beyond [12, 29].

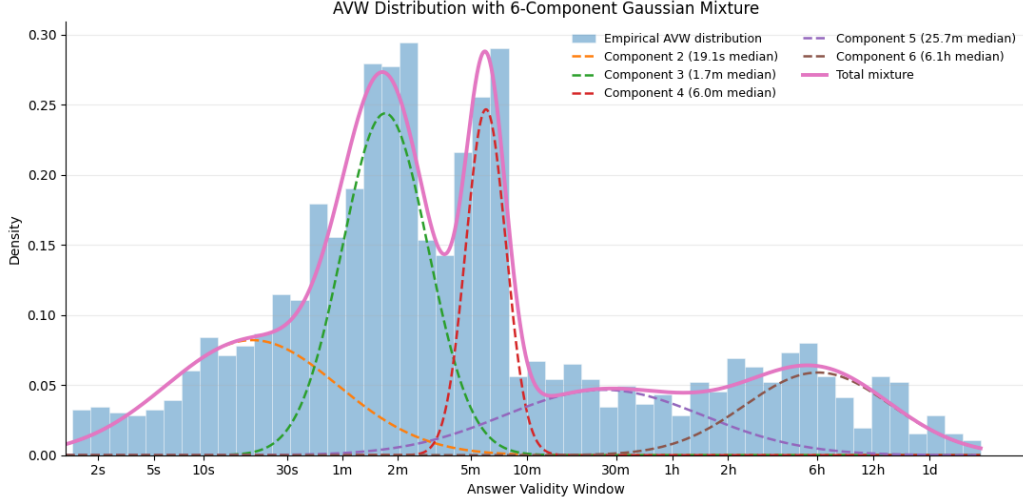


Figure 14: **Answer Validity Window distribution and Gaussian mixture fit.** We plot the empirical distribution of non-zero AVW durations in log-time space and overlay the fitted 6-component Gaussian Mixture Model. The distribution is concentrated around short validity windows but exhibits a long tail of persistent answers, providing empirical support for recall regimes with fine resolution near the evidence moment and broader coverage at longer delays.

Rather than querying each regime at a fixed deterministic offset, we sample recall delays from regime-specific Gaussian distributions centered around representative temporal scales. In particular, we use

Instant recall:	$\Delta t = 0$,
Short-term recall:	$\Delta t \sim \mathcal{N}(2.5, (5/6)^2)$, $\Delta t \in [0, 5]$,
Short-mid-term recall:	$\Delta t \sim \mathcal{N}(17.5, (25/6)^2)$, $\Delta t \in [5, 30]$,
Mid-term recall:	$\Delta t \sim \mathcal{N}(105, 25^2)$, $\Delta t \in [30, 180]$,
Mid-long-term recall:	$\Delta t \sim \mathcal{N}(990, 270^2)$, $\Delta t \in [180, 1800]$,
Long-term recall:	$\Delta t \sim \mathcal{N}(15300, 4500^2)$, $\Delta t \in [1800, 28800]$,
Ultra-long-term recall:	$\Delta t = \text{TTL}$, $\text{TTL} > 28800$.

Samples are accepted only if they fall within the admissible AVW of the corresponding question; otherwise, the regime is not instantiated for that question. This allows the same base question to contribute to multiple recall regimes when its answer remains valid long enough, while short-lived answers contribute only to early regimes.

These regimes should not be interpreted as rigid cognitive boundaries. They are practical evaluation strata: they follow a memory-motivated temporal progression, provide fine resolution near the evidence moment, and become progressively broader at longer horizons.

To verify that this discretization is also compatible with the empirical structure of the data, we fit a Gaussian Mixture Model (GMM) to the non-zero AVW durations in log-time space shown in Figure 14. The distribution is strongly concentrated near short validity windows, but exhibits a long tail of answers that remain valid over minutes, hours, or longer. The fitted components approximately align with the proposed recall regimes, supporting our use of a logarithmic-like scale while leaving room for practical choices due to dataset composition and annotation biases.

A.7 Inference Prompts

We use two prompts during inference, corresponding to the two phases of our streaming protocol: stream ingestion and question answering. The ingestion prompt is used before any question is known, and only serves to initialize the model in a streaming-video setting while incoming frames are encoded into the KV cache. The QA prompt is then appended at the target question time and

queries the model using the visual context currently stored in the cache. This separation prevents question-conditioned video encoding and ensures that all memory policies are evaluated under the same online setting. The ingestion prompt defines the streaming context and marks the beginning of the video stream. No answer is generated during this phase 9. The QA prompt is used after ingestion has reached the desired timestamp. For multiple-choice questions, the model is instructed to output only the selected option index 10.

Prompt 9: Ingestion Phase.

You are experiencing the video from a first-person perspective. Your
 ↳ goal during ingestion is not to write a polished description, but
 ↳ to build a detailed memory that will later be used to answer
 ↳ questions about the video.

Store as much QA-relevant information as possible.

Use first-person narration.

Follow the chronological order of the video.

Do not invent details that are not visible or strongly implied.

Prioritize information useful for future questions:

- Where I am and how the scene is organized.
- What objects are visible and where they are located.
- What objects I interact with.
- Where each object is before I interact with it.
- Where each object is after I interact with it.
- Object attributes such as color, shape, size, material, label,
 ↳ texture, condition, and distinguishing features.
- Object states such as open or closed, full or empty, clean or
 ↳ dirty, broken or intact, switched on or off, covered or
 ↳ uncovered, inside or outside.
- Counts and repetitions, including how many items I see, pick up,
 ↳ place, use, cut, eat, drink, or move.
- The temporal order of events, especially what happens before,
 ↳ during, and after another action.
- People present in the scene, including where they are, what they
 ↳ wear, what they do, and whether I talk to them or work with them.
- Tools, appliances, ingredients, food, containers, drawers, shelves,
 ↳ cabinets, doors, buttons, handles, furniture, vehicles, signs,
 ↳ and landmarks.

For every important object interaction, encode it as a QA memory:

I take [object] from [old location].

I put [object] in/on/under/beside/behind/next to [new location].

I use [tool/object] to [action] [target].

I open or close [object], leaving it [state].

I see [number] [objects] at/in/on [location].

I interact with [person/role] at/in [location].

Avoid vague summaries such as "I prepare food" or "I work on
 ↳ something".

Instead, preserve the concrete objects, locations, quantities,

↳ states, and action order that would allow later question

↳ answering. Remember as much as possible to answer a question.

Prompt 10: Question Answering Phase.

```

Question: [QUESTION]
Options:
1) [OPTION_1]
2) [OPTION_2]
3) [OPTION_3]
4) [OPTION_4]
Reply with number only.

```

A.8 Text Baseline

To assess whether the base QA set can be solved by exploiting textual priors alone, we additionally evaluated a text-only configuration in which the model receives only the question and answer options, without access to the corresponding video (see Figure 7). As shown below, performance is close to random chance across all semantic categories. These results are reported here only as a sanity check and are not directly comparable with the video-based methods discussed in the main paper, since they do not use the visual modality. Nevertheless, they further support the conclusion that the benchmark cannot be solved reliably by leveraging textual bias alone and instead requires visual grounding.

Table 7: Text-only baseline.

Strategy	Configuration	Avg	SPAT	TEMP	DETL	EVNT	SOC	CAUS	PROS
Qwen 8B text only	Text-only MCQA	25.42	23.06	24.06	24.27	27.71	34.63	23.27	20.93

B Implementation Details

B.1 KV-Cache Layout and Streaming Update

Our benchmark uses a unified KV-cache implementation that exposes the same streaming interface for all compression, offloading, and retrieval variants. For each decoder layer, the active cache is stored as a contiguous sequence of three logical regions:

$$K, V = [K_{\text{sink}}, V_{\text{sink}}; K_{\text{mut}}, V_{\text{mut}}; K_{\text{hat}}, V_{\text{hat}}].$$

The sink region contains the priming tokens from the system prompt and initial stream instruction. When sink tokens are enabled, this region is fixed after priming and is never modified by pruning, merging, offloading, or retrieval. The mutable region contains historical visual tokens and is the only region modified by memory-management operations. The hat region contains the most recent visual context and is protected from reduction during streaming. Depending on the configuration, the hat size is either a fixed number of tokens or the token count of the most recent frames.

At each streaming step, a batch of incoming frames is encoded by the visual encoder, projected into the language-model embedding space, and inserted into the language model through the standard image-token placeholders. In frame-aware modes, the cache records the visual-token positions before the forward pass and finalizes the boundary after the reduction step. This removes the auxiliary text/template tokens used to route the images through the processor while preserving the true visual-token frame boundaries.

After each append, a forward pass is done, then the cache checks whether the active length exceeds the configured budget. If it does, the layer first splits the cache into sink, mutable, and hat regions; applies the selected reduction pipeline only to the mutable region; and finally reassembles the cache in the same order. The first decoder layer computes the token indices or trace operations used in a reduction step, and the subsequent layers replay the same operations. This keeps the retained token layout consistent across layers while avoiding independent pruning decisions per layer.

B.2 Memory Budgets and Region Sizes

We separate the active cache into three budgets:

$$B_{\text{active}} = B_{\text{sink}} + B_{\text{mut}} + B_{\text{hat}}.$$

Algorithm 2: Streaming cache update

Input: Incoming frame stream $\{x_t\}$, sink budget B_{sink} , mutable budget B_{mut} , hat budget B_{hat}

Output: Updated active KV cache

Initialize sink cache $K_{\text{sink}}, V_{\text{sink}}$ from the priming prompt

Initialize mutable cache $K_{\text{mut}}, V_{\text{mut}} \leftarrow \emptyset$

Initialize hat cache $K_{\text{hat}}, V_{\text{hat}} \leftarrow \emptyset$

foreach incoming frame batch X_t **do**

 Encode X_t with the visual encoder and projector to obtain visual embeddings

if frame-aware pruning or storage offload is enabled **then**

 Record frame token counts and visual-token positions before the forward pass

 Run the language-model forward pass with the current active cache and the new visual embeddings

 Append the newly produced KV states to the right of the cache during the forward pass

if frame-aware pruning or storage offload is enabled **then**

 Keep only the visual KV states from the newly appended segment

 Update frame-boundary metadata

else

 Update frame-token bookkeeping directly

 Split the resulting cache into sink, mutable, and hat regions

if $|K_{\text{mut}}| > B_{\text{mut}}$ **then**

 Apply the configured mutable-region reduction pipeline

if $|K_{\text{mut}}| > B_{\text{mut}}$ **then**

 Apply the selected overflow policy to the remaining excess mutable tokens

 Reassemble the active cache as $[K_{\text{sink}}, V_{\text{sink}}; K_{\text{mut}}, V_{\text{mut}}; K_{\text{hat}}, V_{\text{hat}}]$

return updated active KV cache

Here, B_{sink} is determined by the number of priming tokens retained as sink tokens, B_{mut} is the live budget for historical visual tokens, and B_{hat} is the protected recent-context budget. The mutable budget is the main control variable for memory compression. All methods compared under the same setting use the same total active-cache budget and the same sink and hat configuration.

When hat tokens are configured by frames, the hat size is computed dynamically from the tracked token counts of the most recent frames. Otherwise, a fixed token-level hat size is used. In both cases, the hat region is excluded from pruning and offloading. If a cache variant does not use frame-aware pruning, the implementation falls back to token-level bookkeeping while preserving the same active-cache accounting.

B.3 Intra-Frame Reducers

Intra-frame reducers operate independently within each mutable frame. This preserves temporal coverage because candidates are generated with frame boundaries and, for token-wise salience methods, the implementation can enforce a minimum number of retained tokens per frame.

Adjacent cosine redundancy (INTRA-COS). For each mutable frame, we compute cosine similarity between adjacent visual-token key states. Candidate adjacent pairs are ranked by similarity. Under pruning, the selected token from each pair is removed. Under merging, the selected token’s value state is averaged into the retained entry. Non-overlap constraints prevent one token from participating in multiple pair reductions in the same pass.

Pseudo-attention salience (INTRA-PA). The query-agnostic pseudo-attention scorer constructs a normalized context vector from the mean key representation of the tokens in a frame. It then scores tokens by their similarity to this context with a softmax-style normalization. Low-importance candidates are selected for reduction.

Value-norm ranking (INTRA-VN). The value-norm reducer scores each visual token by the norm of its value state and removes low-norm tokens. In the current implementation, this reducer is token-wise and supports pruning only. It does not merge low-norm value states into retained tokens.

Sink-guided cross-attention (INTRA-CS). The sink-guided reducer uses the cached sink key states as a fixed guidance template. For each mutable frame, it computes a StreamMem-style attention score between the sink guidance keys and the frame’s visual key states, aggregates the score across heads and guidance tokens using either mean or max reduction, and removes the lowest-scoring visual tokens.

B.4 Inter-Frame Reducers

Inter-frame reducers compare neighboring temporal blocks while preserving the streaming order. The implementation constructs source–reference pairs from consecutive frames in the mutable region and, when a hat region is present, also compares the last mutable frame against the first hat frame. The source side is always a mutable frame, so only mutable tokens are eligible for removal.

Cross-frame cosine redundancy (INTER-COS). For each neighboring frame pair, we compare source-frame visual-token key states against the reference-frame key states. The implementation allows unrestricted matching between tokens across the two frames rather than assuming grid-aligned positions. Each source token receives the maximum cosine similarity to the reference frame, and highly redundant source tokens are selected for removal. A per-source-frame constraint prevents the reducer from removing all tokens from a source frame.

Cross-frame pseudo-attention salience (INTER-PA). For pseudo-attention inter-frame scoring, the reference frame is summarized into a normalized context representation, and source-frame tokens are scored against that context. Tokens with the lowest salience are selected from the mutable source frame. In the current implementation, inter-frame reduction supports pruning only; merging is not used for inter-frame steps.

B.5 Offloading and Quantized Storage

Disk offloading is implemented as an optional storage step after mutable-region reduction. If the mutable region still exceeds B_{mut} , the offload policy removes the leftmost excess mutable tokens and writes them to disk as a contiguous chunk. Each chunk stores keys, values, frame-length metadata, token counts, and bookkeeping fields such as physical bytes and equivalent uncompressed bytes. The offload step may split a frame if only part of that frame lies in the overflow range, and the remaining live frame lengths are updated accordingly.

The offload store supports two formats. In raw mode, keys and values are stored in their original dtype on CPU. In quantized mode, offloaded keys and values are quantized with symmetric per-tensor int8 quantization and saved together with their scales. This quantization affects only the external offload storage. The active cache used during the forward pass remains in the model dtype, and offloaded chunks are dequantized when loaded for retrieval.

B.6 Inference Protocol and Reproducibility

All methods use the same video frame order, frame sampling configuration, query times, system prompt, visual encoder, projector, decoder, and generation interface. During streaming ingestion, position IDs are advanced with a global token counter, while attention masks are constructed from the physical cache length plus the current input length.

For each question, the benchmark clones the current cache state, defined as the master cache, into a secondary cache, identically to the master cache. If retrieval is enabled, relevant blocks are retrieved from the disk and loaded in the secondary cache. The master cache remains unchanged and continues to represent the streaming state of the video, while the secondary cache is used to prompt the model with the question. This design ensures that question-time retrieval does not leak information across questions or alter the cache used for later timestamps.

Reducers, offloading, or retrieval never modify sink and hat regions. Unless otherwise specified, all token-selection scores are computed only over mutable visual tokens. Offloaded tokens are counted separately from live tokens in the reported metrics, and the active cache budget refers to the physical live cache consumed by the decoder at inference time. Retrieval does not increase the active-cache budget; it replaces the historical mutable portion of the active cache with the highest-scoring retrieved blocks under the configured retrieval token budget.

C Experimental Settings

C.1 Hardware and Configuration

All experiments use the same inference setup. Inferences are run on a single NVIDIA A100 GPU with 64 GB of VRAM, 32 CPU cores, and 64 GB of system RAM. Videos are processed at 1 FPS and resized to a target resolution of 896×512 . We use a sliding-window streaming setup with stride 16 and window size 16, corresponding to batches of 16 processed frames, without window overlap. Generation is performed with temperature 0.01 and nucleus sampling with $\text{top-}p = 0.9$. This fixed configuration ensures that all memory-management strategies are compared under identical input resolution, temporal sampling, batching, and decoding conditions.